Journal of Financial Economics 178 (2026) 104237

Contents lists available at ScienceDirect

Journal of Financial Economics

journal homepage: www.elsevier.com/locate/finec

Learning by lending securities☆

George C. Nurisso

Warrington College of Business, University of Florida, Gainesville, FL 32611, USA

ARTICLE INFO

Dataset link: Learning by Lending Securities code (Original data)

JEL classification:
D83
G14
G23

Keywords:
Securities lending
Short-sale constraints
Price efficiency
Information acquisition
Index investing

ABSTRACT

Short sellers convey negative information to securities lenders when borrowing shares. I model how this information generates novel interactions between institutional investors' lending and trading decisions. Lower lending fees improve information quality by facilitating more shorting, but also make it less costly for lenders to strategically recall shares to enhance their trading profits. Lenders may then need to raise fees to commit not to recall shares and thereby attract short sellers. Conversely, index fund lenders cannot trade on lending market information. This restriction enables them to attract greater shorting demand and potentially improve price efficiency—despite charging higher fees.

1. Introduction

Short sellers are informed market participants who must borrow shares to execute their trading strategies. This observation implies that securities lenders can learn valuable information about a stock during the lending process. Specifically, because short sellers are widely regarded as bearers of high-quality negative information, a lender's valuation should drop the moment one borrows shares.[1] Recent empirical work supports this prediction: Honkanen (2025) and Greppmair et al. (2024) show that lenders exploit this signal by reducing their exposure to lent-out stocks. These findings challenge the notion that securities lending is merely a standalone revenue stream for institutional investors, instead revealing that lending and trading are tightly interconnected activities. Consequently, a formal analysis of how these activities interact is needed to understand the implications of these information leakages for financial markets.

In this paper, I model how an institutional investor's ability to obtain information by lending securities affects its lending and trading decisions. This framework helps reconcile recent empirical findings about the securities lending market and provides a new perspective on several key questions in financial markets: how the rise of index investing affects price efficiency, what determines equilibrium lending fees, and which frictions most impede short selling.

I generate three main results. First, lower lending fees improve the precision of the lender's signal. When fees are low, speculators with negative information separate from those with positive information by borrowing shares. This outcome makes the borrowing status of the lender's shares a highly informative trading signal. Second, lenders must raise fees as a commitment device to eliminate their incentive to recall shares. Lenders sell following a loan agreement, which tempts them to recall lent-out shares to boost their expected selling price. Short sellers will not borrow when they anticipate this behavior, so lenders must commit not to by increasing the opportunity cost of recalling shares via higher fees. Third, passive lenders (e.g., index funds) cannot trade on lending market information, which enables them to relax short-selling constraints and, in some circumstances, improve price

☆ Dimitris Papanikolaou was the editor for this article. I thank my advisors Doron Levit and Philip Bond for their invaluable guidance on this project. I would also like to thank Mark Westerfield, Jennifer Koski, Matthew Ringgenberg, Jordan Nickerson, Lukas Kremens, Ahmed Guecioueur, Murillo Campello, and two anonymous referees for insightful comments. I also thank seminar and conference participants at Indiana University, University of Florida, UT Dallas, SMU, Warwick Business School, London School of Economics, Texas A&M, Cleveland Fed, TCU, Seton Hall, University of Washington, and the 2024 Colorado Finance Summit.

E-mail address: george.nurisso@warrington.ufl.edu.

[1] In a survey article on short selling, Reed (2013) refers to short sellers being informed traders as "one of the most robust findings of the literature".

https://doi.org/10.1016/j.jfineco.2026.104237
Received 4 January 2025; Received in revised form 9 January 2026; Accepted 17 January 2026
Available online 27 January 2026
0304-405X/© 2026 Elsevier B.V. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

efficiency. This result occurs even though increased indexing also leads to higher lending fees.

My model involves a securities lender ("he") and an informed short seller ("she") interacting in a lending market before submitting orders to a Kyle (1985) style market maker. Lending fees are endogenous and are chosen to maximize lender profits.[2] Higher lending fees increase revenue per loan but decrease lending volume; less obviously, they also decrease the information conveyed by the lending market. After learning whether the short seller borrowed one of his shares, the lender updates his valuation and decides whether to recall the share (if borrowed) before submitting his order to the market maker. If the lender recalls the share, the short seller cannot sell. The market maker observes the aggregate order flow and sets the price equal to the stock's expected value conditional on that order flow.

I start by considering how the lender's optimal fee choice interacts with his trading strategy. After learning that the short seller borrowed one of his shares, the lender always sells. This trade lowers the short seller's profits by reducing her expected selling price. However, the lender's optimal trading direction conditional on no loan agreement depends on the inference he can make about the stock's payoff given that there was no loan. Intuitively, since borrowing demand is a negative signal, the absence of borrowing demand must be a positive signal. The strength of this inference is decreasing in the fee because when fees are high, the lack of borrowing demand is more likely because of the short seller finding shorting too costly than the stock's payoff being high. Consequently, the lender's optimal fee is decreasing in his planned trading quantity conditional on no loan agreement. When the lender plans on buying following no loan agreement, he favors low fees to ensure that this purchase is well-informed. On the other hand, a lender who always intends to sell prefers high fees to reduce trading competition from short sellers.

I then show that the lender's incentive to recall shares prevents him from having too low a fee. For context, U.S. mutual funds are required by the Investment Company Act of 1940 to be able to recall lent-out shares at will, which, if done, forces the short seller to purchase replacements.[3] As a result, after learning that the stock is overvalued through the lending market, the lender can improve his expected selling price by recalling shares. Chuprinin and Ruf (2018) provide evidence for this practice. The prospect of a higher expected selling price tempts the lender to recall shares, but the short seller anticipates this action and will not borrow if she believes her lender will behave this way. As a result, the lender must commit not to recall shares if he wishes to learn from the lending market: he does so by increasing fees. The lender relinquishes the fee when he recalls shares, so the foregone lending revenue is the cost of this action from his perspective. Thus, the lender loses the incentive to recall shares once the fee is sufficiently high. This no-recall condition establishes a positive lower bound on the set of fees that a lender can charge while still attracting short sellers. The existence of this lower bound is, to my knowledge, new to the short-selling literature and highlights an unintended cost of the regulatory requirement that mutual funds be able to recall shares at will. My analysis shows that this need to commit can considerably increase lending fees and discourage short selling.

The previous two results show that the lender's ability to trade imposes substantial costs on short sellers, so I compare the short-selling constraints that emerge when the lender trades to a benchmark where he cannot. I refer to a lender who cannot trade as passive, which one can interpret as an index fund. Index funds do not have the discretion to trade on information and are prominent players in the securities lending market, making them a natural benchmark.[4] Because they never trade, passive lenders aim to maximize expected lending revenue, which results in them choosing higher fees than active lenders in equilibrium. Nevertheless, passive lenders still attract greater shorting demand because they can inherently commit not to trade on information learned during the lending process or strategically recall shares.

This result also helps reconcile three findings from the growing empirical literature on how passive investing impacts short-sale constraints. First, Honkanen (2025) and Greppmair et al. (2024) show that active securities lenders (i.e., those with the ability to trade) trade on information learned while lending. This result is, of course, part of the motivating evidence for my model. Second, Honkanen (2025), Palia and Sokolinski (2024), and Sikorskaya (2025) find that greater passive ownership of a stock is associated with higher lending fees. Third, von Beschwitz et al. (2025), Palia and Sokolinski (2024), and Sikorskaya (2025) demonstrate that short-selling demand for a stock increases with passive ownership. My model's predictions are consistent with all three findings. In effect, these findings suggest that the information leakage costs imposed by active lenders are more burdensome to short sellers than the higher fees of passive lenders. However, the economic forces that prevent active lenders from setting fees sufficiently low to compensate short sellers for these costs remain unclear. My model provides an answer: the active lender's need to commit not to recall shares prevents him from doing so.

Finally, my model shows that by sufficiently relaxing short-sale constraints, passive lenders can improve price efficiency—a novel result in the theoretical literature on index investing. Empirical evidence by von Beschwitz et al. (2025) and Palia and Sokolinski (2024) supports this prediction. However, even when passive lenders do not improve price efficiency outright, their presence in the lending market still mitigates the damage their lack of informed trading would otherwise cause. Existing theoretical work on index investing typically abstracts from the lending market, often assuming frictionless shorting. My framework therefore suggests that such an abstraction leads to an overestimation of the harm caused by index investing.

1.1. Related literature

To my knowledge, only two other papers explicitly model information spillovers during the securities lending process: independent and contemporaneous works by Pankratov (2020) and Chen et al. (2025).[5]

Pankratov (2020) numerically analyzes a model of endogenous short-sale constraints in which security lenders balance lending revenue against information when setting fees. In contrast, my paper contributes methodologically by developing a tractable theoretical framework for analyzing the equilibrium consequences of information leakages in the securities lending market. This framework admits closed-form solutions, enabling a more transparent characterization of the underlying economic forces while maintaining the flexibility to incorporate key institutional details. For example, it provides the first theoretical analysis of strategic share recall, showing how this possibility imposes a positive lower bound on the fees active lenders can charge. This insight, in turn, implies that index funds' built-in commitment not to recall shares or trade on lending market information gives them an inherent advantage in attracting short sellers.

Chen et al. (2025) study market power in securities lending and find evidence for market concentration and non-competitive fees. The

[2] One can interpret this process as either the lender directly setting fees or an unmodeled lending agent setting fees on the lender's behalf.

[3] https://www.sec.gov/divisions/investment/noaction/1972/statestreet052272.pdf The implication is that lenders covered by the Investment Company Act of 1940 (e.g., mutual funds) must have this provision in their lending agreements. Engelberg et al. (2018) document recall risk as a unique risk faced by short sellers.

[4] Blocher and Whaley (2016) document that lending revenue can exceed management fees for indexed ETFs, and Chinco and Sammon (2024) estimate that passive investing comprises over one-third of the US stock market.

[5] Baldauf and Mollner (2024) model information leaks in a more general over-the-counter setting. See Duffie et al. (2002); Blocher et al. (2013); Atmaz et al. (2023); Johnson and Weitzner (2025); Mainardi (2023); and Nezafat and Schroder (2021) for models of securities lending without information spillovers.

authors then propose a model to explain why such market power exists. They compare two different lending market structures: an over-the-counter setting where lenders with market power set fees to maximize lending revenue (similar to my passive lender case) and a competitive, centralized market where borrowing demand is observed by all participants. In the former setting, we see elevated fees because of the lender's market power, but no information leakages because the lender does not trade. In the latter, competition drives fees to zero, but short sellers are worse off because their borrowing demand is immediately revealed to all. A fundamental assumption in this model is that lenders in the over-the-counter setting do not trade on whatever information they learn from the lending market. My paper relaxes this assumption and endogenizes the quality of that information through the lender's fee choice. These changes are not just theoretical exercises: they reflect the observed behavior of securities lenders in recent empirical research. Honkanen (2025) and Greppmair et al. (2024) demonstrate that active lenders rebalance their portfolios away from stocks on loan and find that the associated sales precede further price declines. Similarly, Chague et al. (2025) show that Brazilian brokers leak lending market information to other clients.

Contemporaneous work by Sikorskaya (2025) also examines how the rise of passive investing affects shorting demand, albeit through a different channel. In her model, asset managers optimally tilt their portfolios toward the index against which they are benchmarked. This additional institutional demand for index assets increases both lendable supply (via greater institutional ownership) and shorting demand (since the index assets become overvalued), making the equilibrium effect on borrowing fees ambiguous. Exploiting Russell 1000/2000 index reconstitutions, Sikorskaya (2025) provides empirical evidence that the demand effect prevails: greater benchmark-driven ownership leads to higher lending fees. This benchmarking channel complements the information leakage channel explored in this paper as a mechanism through which indexing can lead to greater shorting demand. Nevertheless, von Beschwitz et al. (2025) show that stocks with negative post-inclusion returns still experience increases in shorting demand following index inclusion (with magnitudes nearly as large as those with positive returns), highlighting the robustness of the information leakage channel.

This paper also contributes to the theoretical literature on how passive investing impacts price efficiency. There is no overwhelming consensus in this literature. For example, Bond and Garcia (2022) find that passive investors can improve price efficiency, while Coles et al. (2022) find that passive investors have no effect. On the other hand, Baruch and Zhang (2022) and Nurisso (2025) solve models where increased indexing harms price efficiency. However, none of the above papers explicitly analyze how the growth in passive investing impacts price efficiency through the securities lending market. I contribute to this literature by generating the novel theoretical result that passive funds can improve price efficiency by relaxing short-sale constraints. von Beschwitz et al. (2025) and Palia and Sokolinski (2024) empirically show that passive lenders can improve the ability of prices to incorporate negative information through this channel. However, it is unclear whether passive lenders increase overall price efficiency if they facilitate the incorporation of negative information but hinder the incorporation of positive information. My model provides conditions under which passive lenders can improve overall price efficiency by relaxing short-sale constraints.

The paper proceeds as follows. Section 2 outlines my model. Section 3 solves for the equilibrium. Section 4 examines various extensions: considering the lender's reputation as a commitment device, allowing the lender to sell recalled shares, making the lender informed, and introducing lender competition. Section 5 discusses the empirical predictions and policy implications of my model. I then conclude.

2. Model setup

The model contains four agents: an informed speculator (S), a lender (L), noise traders, and a market maker. All agents are risk-neutral. There is a single firm whose stock is traded in a financial market. The firm's payoff is $v \in \{0,1\}$ where $Pr(v = 1) = \rho \in (0,1)$. I analyze a static model with two main phases: lending and trading. In the lending phase, the speculator can borrow shares from the lender for a fee f. Next, the lender, the speculator, and the noise traders submit their asset demands to the market maker, who sets the stock's price $P(z)$ equal to its expected payoff conditional on the aggregate order flow z. After the price is generated, the firm's payoff v is revealed. I assume the speculator is perfectly informed about the realization of v while the lender is uninformed.[6]

The speculator faces a random opportunity cost of trading, c, that is uniformly distributed over $[0, \bar{C}_v]$. In general, this distribution can depend on the underlying firm value v. The realization of c is the speculator's private information. The speculator only attempts to trade if her expected profits exceed c. One can interpret this situation as the speculator having a finite amount of capital that she can deploy in this period, with c representing the profits of her next best option.[7]

2.1. Lending and trading

I assume the lender L owns enough shares to lend without constraining his subsequent ability to trade[8]; this assumption implies that the lender owns $N \geq 2$ shares of the stock.[9] The lender's asset demand will be $D_L \in \{-1, 0, 1\}$ where 1 represents a purchase, 0 represents no trade, and -1 represents a sale (not a short sale).[10] Passive lenders are a special case where D_L always equals zero. On the other hand, the speculator S owns zero shares, so she must borrow shares from the lender to sell the stock short. Her demand is: $D_S \in \{-1, 0, 1\}$, but $D_S = -1$ signifies a short sale. The speculator must therefore first borrow a share from the lender to have $D_S = -1$.

The lender chooses a fee $f \geq 0$ to maximize his expected profits. Note that fees will always be non-negative: negative fees amount to the lender paying the speculator to borrow his shares, and this choice is never optimal for the lender.[11] This fee is a "take-it-or-leave-it" offer sent to the speculator. The speculator borrows if and only if her

[6] I discuss the case where the lender also has perfect information in Section 4.3 and the case where the lender has an imperfect private signal Online Appendix E.

[7] In Online Appendix C, I discuss an alternative modeling approach in which the speculator's information quality is a random variable and her opportunity cost is constant. My main results continue to hold under this alternative approach.

[8] This assumption is motivated by existing regulations that prohibit mutual funds from lending more than one-third of their total asset value. Furthermore, Sikorskaya (2025) shows that mutual fund families often voluntarily restrict securities lending by setting position-level caps below 100%.

[9] This analysis takes the lender's ex-ante portfolio as given, thereby abstracting from any initial portfolio differences between active and passive lenders. However, this distinction is less relevant when the active lender is uninformed, as he may behave as a "closet indexer" and hold a portfolio close to that of the passive lender (see Brown and Davies, 2017).

[10] I assume the lender cannot sell recalled shares. When a lender recalls shares, the borrower returns them on the standard settlement cycle. Since trade occurs immediately after the lender receives his information (before recalled shares are returned), selling recalled shares would expose the lender to settlement risk, as delivery depends on the borrower returning them on time. I relax this assumption in Section 4.2.

[11] With a negative fee, it is always optimal for the speculator to borrow, regardless of the realization of v. This outcome destroys the informativeness of the lending market signal and produces negative lending revenue. Thus, a negative fee can never be optimal.

expected selling profits minus f exceed her cost c.[12] The lender's fee choice resembles a monopolist selecting the profit-maximizing point on a downward-sloping demand curve.[13] All else equal, a lower fee induces a broader range of speculators (i.e., those with higher values of c) to sell the stock short when $v = 0$.

After observing whether the speculator borrowed a share, the lender rationally updates his stock valuation. Denote the lending market signal as $b \in \{1, 0\}$ where $b = 1$ represents the outcome where the speculator borrows the lender's share and $b = 0$ represents the outcome where the speculator does not borrow the lender's share. Motivated by existing regulations, the lender can also recall the lent-out share between the lending and trading stages. If the lender recalls the share, he returns the fee to speculator and the speculator can no longer sell. This outcome implies that the speculator would make zero trading profits, so her total profits net of opportunity cost c would be negative. Thus, the speculator will not attempt to borrow if she expects her lender to recall the loan.

The financial market is a discrete version of Kyle (1985). Besides the speculator and the lender, there is also noise trade. Aggregate noise demand takes values: $D_{NT} \in \{-2, -1, 0, 1, 2\}$ with each outcome having equal probability. I assume that the realization of noise demand is independent of all other random variables and that negative noise demand corresponds to *sales* and not short sales—only the speculator shorts.[14] Noise traders also never lend.

All traders submit their demands to a risk-neutral, competitive market maker. The market maker observes the aggregate order flow

$$z \equiv D_S + D_L + D_{NT}$$

and sets the price equal to the expected value of v conditional on that order flow z. Thus, I define the price as:

$$P(z) \equiv E[v|z] \tag{1}$$

Importantly, the market maker does not observe developments in the lending market (both fees and quantities) and, therefore, cannot distinguish regular sales from the more informative short sales. This inability to observe lending market outcomes allows the lender to trade profitably on the lending market signal b.[15] After the payoff v is revealed, all short sales are completed by the speculator paying her lender the terminal payoff v.

2.2. Timing

The timing of this game is as follows:

1. The payoff v is generated.
2. The speculator learns the realizations of v and her cost c.
3. **Lending Market**
   (a) The lender chooses a borrowing fee $f \geq 0$ and offers it to the speculator.
   (b) The speculator decides whether to borrow a share at fee f.
   (c) The lender learns whether his share was borrowed and updates his valuation of the asset.
   (d) The lender decides whether to recall the loan. If so, the speculator must return the share and the lender must return the fee f. If recalled, the speculator can no longer sell.
4. All traders submit their asset demands to the market maker. The price $P(z)$ is then generated through the pricing rule (1).
5. The realization of v is made public. All short sales are completed by the borrower paying the lender v.[16]

2.3. Equilibrium concept

My solution concept for this model is the Perfect Bayesian Equilibrium (PBE). An equilibrium requires (i) a trading strategy for the speculator that maximizes expected profits given the price-setting rule, the fee, the lender's equilibrium trading and recall strategies, and the realizations of the firm's payoff v and her cost c; (ii) a fee-setting, trading, and recall strategy for the lender that maximizes expected profits given the price-setting rule, the lending market signal b, and the speculator's equilibrium trading strategy; and (iii) a price-setting rule for the market maker that follows Eq. (1) given the strategies of the speculator and the lender. The speculator, the lender, and the market maker have rational expectations in that they update probabilities via Bayes' rule and have correct-in-equilibrium beliefs about the other players' strategies.

One can view the market maker in my model as the "receiver" in a multi-sender signaling game, so I will apply the notion of "unprejudiced beliefs" from (Bagwell and Ramey, 1991) as my primary equilibrium refinement. This refinement requires the receiver to believe that off-equilibrium outcomes come from the fewest number of deviations from equilibrium play. In my model, a market maker with unprejudiced beliefs should view an off-equilibrium order flow as coming from a single trading deviation by the speculator *or* the lender instead of by both simultaneously.[17]

3. Model solution

I describe my solution procedure in Section 3.1. In Section 3.2, I solve for the equilibrium with a passive lender under the assumption that the lender cannot recall shares. In Section 3.3, I find all equilibria with an active lender under the assumption that the lender cannot recall shares. In Section 3.4, I analyze how granting the lender the ability to recall shares changes the equilibria. I then compare the various equilibria and examine the implications for price efficiency in Section 3.5.

3.1. Solution procedure

I solve the model under the following parametric assumptions.

Assumption 1. $\bar{C}_1$ is low enough to ensure that the speculator always buys when she learns that $v = 1$.

[12] In the base model, I rule out "manipulative" borrowing strategies to focus on the fundamental case where informed traders only borrow shares to short sell. See Online Appendix B for an analysis of manipulative borrowing.

[13] There is strong evidence that securities lenders have market power. See Duffie et al. (2002), Kolasinski et al. (2013), and Chen et al. (2025). I discuss competition between lenders Section 4.4

[14] Allowing noisy short sales would make $b = 1$ a less negative signal about firm value. My model, therefore, analyzes the extreme case where short-selling demand is perfectly informative about $v = 0$ rather than just highly informative. Note also that securities lenders seem able to identify informative short sales, which weakens the strength of this assumption (see Honkanen, 2025).

[15] This assumption is justified as long as there is some gap in time between when the lender and borrower complete the loan and when this information becomes publicly available. If the market maker could observe real-time borrowing demand, the price would immediately reflect the short seller's information, destroying her incentive to trade.

[16] The fact that the lender is perfectly informed and returns the share at fundamental value implies zero credit risk. In practice, credit risk is usually minimal since the loans are over-collateralized and lenders receive various protections from intermediaries.

[17] Given the structure of noise demand, off-equilibrium order flows will only be at the endpoints of the set of feasible order flows: $z = 4$ or $z = -4$. At these potential off-equilibrium order flows, the speculator must either be buying or shorting, respectively, so unprejudiced beliefs are sufficient to pin down the price as $P(4) = 1$ and $P(-4) = 0$.

Assumption 2. $\bar{C}_0$ is the maximum opportunity cost that ensures the speculator always sells when $v = 0$ in the counterfactual setting where she already owns a share (and thus never needs to borrow).

Assumption 1 says that the exogenous cost of trade c is never high enough to deter the speculator from buying when $v = 1$. This assumption allows me to focus on how the lending market impacts short selling and contrast those results with "frictionless" buying. As such, $\bar{C}_1$, the maximum opportunity cost when $v = 1$, does not play a role in the main analysis. Assumption 2 then facilitates a more exact understanding of how the lending market impacts outcomes by establishing a clear counterfactual benchmark: when the speculator does not need to borrow shares to sell, the unique equilibrium has the speculator always buy when $v = 1$ and always sell when $v = 0$.[18] The exogenous cost c alone does not deter the speculator's trades: it only does so when combined with lending market frictions. My main results do not depend on these specific parameter restrictions, but they make the model more tractable and easier to interpret.[19] Given these assumptions, I will hereafter refer to $\bar{C}_0$ as $\bar{C}$ to minimize notational clutter.

I conjecture that the speculator's trading strategy conditional on $v = 0$ follows a threshold rule. The speculator will short sell when $v = 0$ if and only if $c < c^*$ where $c^* \in [0, \bar{C}]$.[20] The lender's trading strategy will depend on whether the speculator borrowed one of his shares (i.e., the realization of the lending market signal b). Denote $D_L(b) \in \{-1, 0, 1\}$ as the lender's demand when he observes the lending market signal b.

Lemma 1. *The lender always sells when the speculator borrows a share. That is,* $D_L^*(b = 1) = -1$.

The proof of Lemma 1 is straightforward. The speculator only sells short when she knows $v = 0$, so the lender's valuation also becomes zero after making a loan. Thus, selling is always optimal for the lender since prices can never go below zero. However, the lender's optimal trading strategy following no loan agreement (i.e., the lending market signal $b = 0$) is less obvious. I focus on pure strategies for the lender's trades, so $D_L(b = 0) \in \{-1, 0, 1\}$. I also assume the convention that the lender will not trade if indifferent about doing so for the rest of the paper.

I solve for all equilibria by guessing and verifying whether a particular lender demand conditional on no loan agreement ($D_L(b = 0)$) can be part of the lender's equilibrium trading strategy. I then repeat this process for all possible values of $D_L(b = 0)$.

3.2. Passive lender

I first solve the model in the benchmark case where the lender is passive and not allowed to recall shares. The passive lender's defining characteristic is that he does not trade, so D_L always equals zero (regardless of the realization of the lending market signal b). One can interpret the passive lender as an index fund that cannot deviate from its index.

I solve the model backward. First, given the speculator's conjectured trading strategy and the fact that the passive lender never trades, the market maker generates prices $P(z) = Pr[v = 1|z]$, where z is an integer between negative three and three. The prices are functions of the equilibrium short-selling threshold c^*. Specifically,

$$P(z) = \begin{cases} 1 \text{ when } z = 3 \\ \frac{\rho}{\rho + (1-\rho)(1-G(c^*))} \text{ when } z = 2 \\ \rho \text{ when } z \in \{1, 0, -1\} \\ 0 \text{ when } z \in \{-2, -3\} \end{cases} \tag{2}$$

where $G(x) \equiv \frac{x}{\bar{C}}$ is the cumulative distribution function of the opportunity cost c when $v = 0$. Intuitively, $P(z) = 1$ means order flow z only occurs when $v = 1$ and $P(z) = 0$ similarly means that z only appears when $v = 0$. $P(z) = \rho$ implies that z is an uninformative order flow (i.e., can always appear in equilibrium), and $P(z) = \frac{\rho}{\rho + (1-\rho)(1-G(c^*))}$ corresponds to a z that only appears when the speculator does not short sell; that is, when $v = 1$, or $v = 0$ and $c \geq c^*$. Observe that since the prices are functions of conjectured equilibrium strategies, the lender and the speculator take them as given.

I must now derive the equilibrium cutoff c^*. Given the lender's choice of fee f, the *actual* short-selling threshold (which I denote as $c(f)$) must satisfy

$$c(f) = E[P(z)|D_S + D_L = -1] - f \tag{3}$$

All speculators with $c < c(f)$ will then find it optimal to short when $v = 0$, and all speculators with $c \geq c(f)$ will find it optimal not to trade when $v = 0$.

Next, I solve for the passive lender's optimal fee. Since the passive lender never trades, he chooses the fee to maximize his expected lending revenue.

$$\max_f Pr[D_S = -1]f = Pr[v = 0]\left(\frac{E[P(z)|D_S + D_L = -1] - f}{\bar{C}}\right) f \tag{4}$$

where $Pr[v = 0]$ equals $1 - \rho$. Also observe that

$$E[P(z)|D_S + D_L = -1] = \frac{1}{5}\sum_{z=-3}^{1} P(z)$$

Solving (4) generates the passive lender's optimal fee.

$$f^* = \frac{E[P(z)|D_S + D_L = -1]}{2} = \frac{3\rho}{10} \tag{5}$$

Eq. (5) says the passive lender optimally collects half the short seller's expected selling profits as a fee. Moreover, this fee is increasing in the ex-ante probability that the payoff equals one. Intuitively, when v is expected to equal one with high probability (i.e, ρ is close to one), short sales are unlikely but extremely profitable when v does in fact equal zero. The passive lender then raises his fee accordingly to maximize the revenue that he can collect from this rare but lucrative event.

I then pin down the equilibrium shorting threshold c^* by substituting (5) into Eq. (3), setting $c(f) = c^*$, and solving for c^*. Proposition 1 summarizes the findings for the passive lender.

Proposition 1. *The passive lender's optimal fee maximizes expected lending revenue and takes half of the short seller's expected profits. The resulting equilibrium fee is described by Eq. (5).*

Proposition 1 describes the outcome with a lender who sets fees to maximize expected lending revenue and never trades. The existing theoretical literature on securities lending typically assumes this sort of lender behavior. I now analyze what happens to these outcomes when the lender trades on lending market information.

3.3. Active lender

The solution procedure is quite similar when the lender trades. The only difference is that I now allow the lender to trade optimally instead of imposing $D_L = 0$. That is, the market maker believes that the lender sells following a loan agreement (see Lemma 1) and demands $D_L^*(b = 0) \in \{-1, 0, 1\}$ following no loan agreement. The market maker

[18] See Proposition S1 in the appendix for more details.

[19] See Online Appendix A for an analysis of what happens when these assumptions are relaxed.

[20] Assumption 2 implies that the imposition of any costs on short sellers (either direct via fees or indirect via information leakage) will result in an equilibrium $c^* < \bar{C}$, allowing us to focus on the more relevant interior case where shorting demand is cost-sensitive.

also continues to believe that the speculator always buys when $v = 1$ and will borrow and sell short when $v = 0$ if and only if $c < c^*$. The market maker then sets prices given these conjectures.

First, I must solve for the equilibrium short-selling cutoff given the lender's conjectured trading strategy. Given any fee choice f, the *actual* cutoff is:

$$c(f) = E[P(z)|D_S + D_L = -2] - f \tag{6}$$

Eq. (6) only differs from its passive counterpart (3) in that the expected selling price is now conditional on $D_S + D_L = -2$ rather than -1. This change follows from the speculator's belief that the lender always sells after making a loan agreement (see Lemma 1).

I next examine the lender's stock valuation after lending at some arbitrary fee f.

$$E[v|b = 1] = 0 \tag{7}$$

$$E[v|b = 0] = \frac{\rho}{\rho + (1-\rho)(1 - G(c(f)))} \tag{8}$$

Since the short seller is perfectly informed, a loan agreement (i.e., $b = 1$) implies $v = 0$. It is always optimal for the lender to sell in this scenario because prices are guaranteed to be no lower than zero. On the other hand, no loan agreement (i.e., $b = 0$) could occur because the speculator knows that $v = 1$ or because the speculator knows that $v = 0$ but has a high realization of c. The lender's valuation (8) is a weighted average of these two outcomes.

Lemma 2. *The lender's valuation of the asset conditional on no loan agreement is decreasing in f.*

Higher fees make the speculator less inclined to short the stock. This response makes the absence of a loan agreement more likely to reflect the case where the speculator knows $v = 0$ but finds shorting too costly, thereby lowering the lender's valuation.

Moving back to the fee-setting stage, the lender's objective is to maximize his expected lending plus trading profits.

$$\max_f Pr[D_S = -1]\Big(E[P(z)|D_S + D_L = -2] + f\Big) + \Big(1 - Pr[D_S = -1]\Big)\Big(D_L(b=0)E[v - P(z)|b = 0, D_L(b=0)]\Big) \tag{9}$$

The first term in Eq. (9) corresponds to the event where the speculator borrows a share. Conditional on this event, the lender's profits equal the fee f plus his expected selling profits $E[P(z)|D_S + D_L = -2] - 0$. The second term corresponds to the event where the speculator does not borrow a share. The lender's profits conditional on this event equal his expected trading profits: $D_L(b=0)E[v - P(z)|b = 0, D_L(b=0)]$. Eq. (9) furthermore tells us that a higher fee has three effects on the lender's expected profits. First, there is the direct effect of a higher fee leading to greater lending revenue conditional on a loan. Second, a higher fee reduces the likelihood of a short sale when $v = 0$; that is, $Pr[D_S = -1] = (1-\rho)\frac{E[P(z)|D_S+D_L=-2]-f}{\bar{C}}$ is decreasing in f. Third, as Lemma 2 states, higher fees decrease $E[v|b = 0]$. This third effect implies that the lender's trading profits conditional on no loan agreement are decreasing in f when he buys, increasing in f when he sells, and are unaffected by f when he does not trade. I characterize the optimal fees as a function of each possible trading strategy $D_L(b = 0) \in \{-1, 0, 1\}$ below.

Proposition 2. *Given some set of prices $P(z) \in [0, 1]$ for $z \in \{-4, \dots, 4\}$, each possible trading strategy conditional on no loan agreement ($D_L(b = 0) \in \{-1, 0, 1\}$) maps to a unique optimal fee $f^* \geq 0$. Furthermore, the fee associated with buying ($D_L(b = 0) = 1$) is no higher than the fee associated with not trading ($D_L(b = 0) = 0$), and the fee associated with not trading ($D_L(b = 0) = 0$) is strictly lower than the fee associated with selling ($D_L(b = 0) = -1$).*

Specifically, these fees are:

$$f^* = \begin{cases} \max\left(\frac{-E[P(z)|D_S+D_L=1]}{2}, 0\right) = 0 \ when \ D_L(b=0) = 1 \\ 0 \ when \ D_L(b=0) = 0 \\ \max\left(\frac{E[P(z)|D_S+D_L=-1]}{2}, 0\right) = \frac{E[P(z)|D_S+D_L=-1]}{2} \ when \ D_L(b=0) = -1 \end{cases} \tag{10}$$

Observe that because the active lender always sells following a loan agreement (Lemma 1), any differences in the active lender's optimal fee must come from the second term in (9): the active lender's expected profits conditional on no loan agreement. As a baseline, this term equals zero when the lender does not trade following no loan agreement (i.e., $D_L(b = 0) = 0$). When the lender buys following no loan agreement (i.e., $D_L(b = 0) = 1$), this second term is strictly decreasing in f, so his optimal fee is pushed downward relative to the case where he does not trade. Intuitively, a lower fee raises the lender's valuation conditional on no loan agreement, since this outcome now more likely reflects a speculator who knows $v = 1$ (see Lemma 2). The lender, therefore, subsidizes short selling to improve the quality of his lending market signal.[21] Conversely, the lender's profits conditional on no loan agreement are strictly increasing in f when he sells, pushing his optimal fee upwards relative to the no-trade case. By setting a higher fee, the lender can kick short sellers out of the market and generate a higher expected selling price ($E[P(z)|D_S + D_L = -1]$ rather than $E[P(z)|D_S + D_L = -2]$) for himself.[22]

We now have the optimal fee for any given lender trading strategy. To verify a candidate equilibrium trade, I plug the associated fee into Eq. (6), set $c^* = c(f)$, and solve the resulting fixed-point problem for c^*. Once I find the solution, I check whether the conjectured trade is indeed optimal. First, I verify that the lender has no incentive to deviate to another trading strategy while holding the fee fixed. Then, I confirm that the lender does not prefer any alternative fee–trading strategy pair. The following proposition characterizes the equilibria with an active lender.

Proposition 3. *The active lender always sells after a share is borrowed. Both buying and not trading after no loan agreement are equilibrium trades ($D_L^*(b = 0) \in \{0, 1\}$). Selling following no loan agreement is never an equilibrium trade for the active lender. Equilibrium fees equal zero.*

Equilibrium fees follow from the lender's equilibrium trading strategies and Eq. (10).[23] The lender's optimal trade following a loan agreement is a consequence of Lemma 1. The lender's optimal trade following no loan agreement depends on his updated valuation of the stock, given by Eq. (8). Observe that this valuation exceeds the lender's prior belief that $v = 1$ (i.e., ρ), so the absence of a loan agreement ($b = 0$) is a positive signal. It then follows that selling conditional on $b = 0$ is never optimal. Thus, the lender will either buy or not trade depending on whether this valuation exceeds his expected buying price. In fact, Proposition 3 establishes that both buying and not trading are equilibrium strategies. The multiplicity of equilibria arises from how prices respond to the lender's trades. To make this connection more transparent, I re-express the equilibrium price as a function of both the

[21] Of course, how far this fee can be lowered is constrained by the zero lower bound on lending fees.

[22] Note that the prices are non-decreasing in the order flow, as is typical in Kyle (1985) style microstructure models.

[23] The result that the active lender's equilibrium fee equals zero depends on the assumption that c is uniformly distributed. However, the broader insight that the lender wants to lower fees below the revenue-maximizing level to learn holds for any distribution of trading costs where $G'(f) < 0$; that is, where higher fees reduce the probability of a loan.

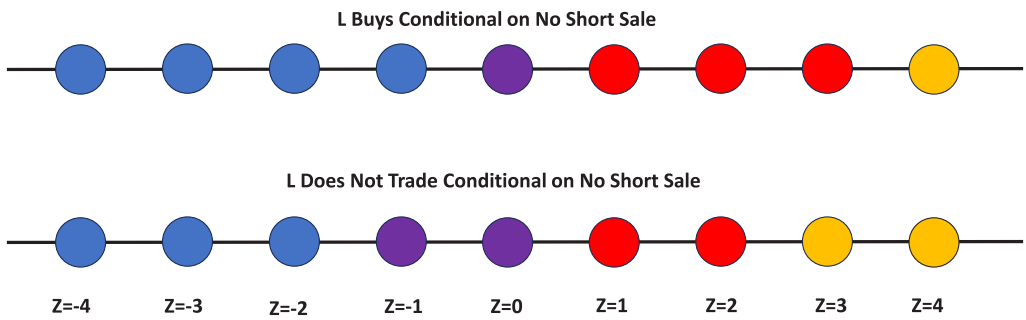

Fig. 1. Equilibrium Prices.
$P(z)$ for different trading strategies conditional on $b = 0$ for the lender L. Yellow dots represent $P(z) = 1$, red dots represent $P(z) = \frac{\rho}{\rho+(1-\rho)(1-G(c^*))}$, purple dots represent $P(z) = \rho$, and blue dots represent $P(z) = 0$.

order flow and the lender's equilibrium trading strategy conditional on no loan agreement: $P(z, D_L^*(b = 0))$.

Corollary 1. *For all order flows $z \in \{-4, 4\}$, equilibrium prices $P(z, D_L^*(b = 0))$ are weakly decreasing in the lender's equilibrium trade conditional on no loan agreement $D_L^*(b = 0)$. Moreover, expected prices conditional on the sum of the speculator's and the lender's trades (i.e., $E[P(z, D_L^*(b = 0))|D_S+D_L]$) are strictly decreasing in $D_L^*(b = 0)$ for all $D_S+D_L \in \{-2, \dots, 2\}$.*

Corollary 1 is a comparative static of the equilibrium prices derived in the proof of Proposition 3. Fig. 1 provides intuition for this corollary by characterizing the mappings between prices and order flows in the two possible equilibria: the lender buying (top row) and not trading (bottom row) conditional on no loan agreement. Yellow dots correspond to a price of 1, red dots to a price of $\frac{\rho}{\rho+(1-\rho)(1-G(c^*))} \in [\rho, 1]$, purple dots to a price of ρ, and blue dots to a price of 0.[24] As the lender's demand decreases from one to zero, the set of order flows that can appear when $\upsilon = 1$ shifts downward. This outcome generates greater overlap between the sets of order flows that can occur when $\upsilon = 0$ and $\upsilon = 1$. As a result, order flows that previously *only* occurred when $\upsilon = 0$ can now also appear when $\upsilon = 1$, increasing the prices associated with those order flows.[25] Consequently, the lender's expected *equilibrium* buying price is lower than his expected *deviation* buying price when the market maker believes that he is not trading. Thus, the lender will buy when the market maker expects him to and not trade when the market maker instead has that belief, generating multiple equilibria.

3.4. Recall risk

I now allow lenders to recall shares as per the timeline given in Section 2.2. This feature is motivated by existing regulations that require all investment companies covered by the 1940 Investment Company Act (e.g., U.S. mutual funds) to be able to recall lent-out shares at will. To summarize my modeling of this regulation, the lender can recall the lent-out share after he and the borrower reach a loan agreement, but before they submit their demands to the market maker. If the lender recalls the loan, the lender loses the fee, and the speculator cannot sell short. In actual lending markets, the lender would recall the loan after the speculator sold short, forcing her to buy. One can, therefore, equivalently interpret my timeline as the speculator submitting a sell order followed by a buy order when recalled, and the market maker only observing the net impact of the speculator's trades on the order flow: zero.

First, the passive lender never wants to recall shares: doing so entails a loss of fee income without any benefit. The active lender, however, may want to. More specifically, conditional on lending a share, an active lender will not want to recall it if and only if the following condition on fees is satisfied:

$$E[P(z)|D_S + D_L = -2] + f \geq E[P(z)|D_S + D_L = -1]$$
$$\implies f \geq \Delta E[P(z)] \equiv E[P(z)|D_S + D_L = -1] - E[P(z)|D_S + D_L = -2] \tag{11}$$

By not recalling, the lender expects to sell at $E[P(z)|D_S + D_L = -2]$ (the expected price conditional on the speculator and lender selling) and collect the fee f. By recalling, the lender loses the fee but expects to sell at the higher price $E[P(z)|D_S + D_L = -1]$ (the expected price conditional on the lender selling and the speculator not trading). Note that $\Delta E[P(z)] = \frac{1}{5}(P(1) - P(-4)) > 0$, so $f = 0$ does not satisfy the no-recall condition given by (11). As a result, the lender cannot commit not to recall shares at the equilibrium fee from Section 3.3 ($f^* = 0$), meaning that the speculator will now never borrow from the lender at that fee.[26] Thus, if the lender wants to attract short sellers, he must raise his fee to satisfy (11). Fig. 2 illustrates this intuition by plotting the probability of a short sale when $\upsilon = 0$ against the lending fee. We can interpret the blue line in Fig. 2 as the lending market's demand curve, with the probability of a short sale when $\upsilon = 0$ representing this paper's notion of quantity demanded. We obtain a standard downward-sloping demand curve above the $\Delta E[P(z)]$ lower bound: lower fees attract greater shorting demand. However, once fees fall below the lower bound, the probability of a short sale collapses to zero because the lender can no longer commit not to recall shares.

The following lemma characterizes how the active lender's optimal fee changes with recall risk.

Lemma 3. *The active lender's optimal fee without recall risk (i.e., Eq. (10) from Section 3.3) violates the no-recall condition if and only if he buys or does not trade conditional on no loan agreement. If the lender's optimal fee without recall risk violates the no-recall condition, his new optimal fee with recall risk becomes $\Delta E[P(z)]$. Otherwise, his optimal fee does not change.*

The lender's optimal fee without recall risk violates the no-recall condition when he buys or does not trade conditional on no loan agreement because $\Delta E[P(z)] > 0$. As a result, the lender will select the lowest fee satisfying (11) when recall risk is present: $f^* = \Delta E[P(z)]$. Intuitively, the optimal fee without recall risk from Section 3.3 maximized

[24] Observe that $z = 4$ does not occur in equilibrium when $D_L^*(b = 0) = 0$, but the unprejudiced beliefs refinement (see Section 2.3) implies that $P(4) = 1$ in this scenario.

[25] The only prices that depend on the equilibrium value c^* are those represented by the red dots; that is, when $P(z, D_L^*(b = 0)) = \frac{\rho}{\rho+(1-\rho)(1-G(c^*))}$. Within these order flows, equilibrium prices are still higher when $D_L^*(b = 0) = 0$ than when $D_L^*(b = 0) = 1$ since $G(c^*)$ will be greater under the former. See the proof for more details.

[26] The speculator earns zero trading profits when recalled, which is never above her opportunity cost $c \in [0, \bar{C}]$.

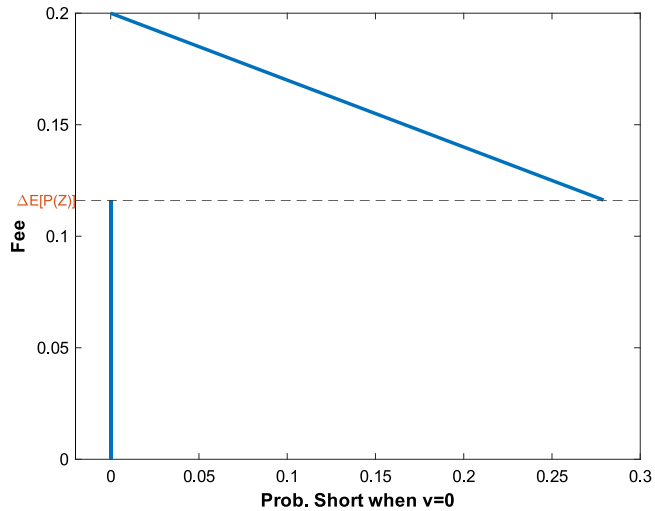

Fig. 2. Lending Market Demand Curve.
Fig. 2 plots the probability of a short sale when $v = 0$ (i.e., $G(c(f))$) against the fee for an active lender who sells following a loan agreement and does not trade otherwise. I also set $\rho = 0.5$ and $\bar{C} = 0.3$.

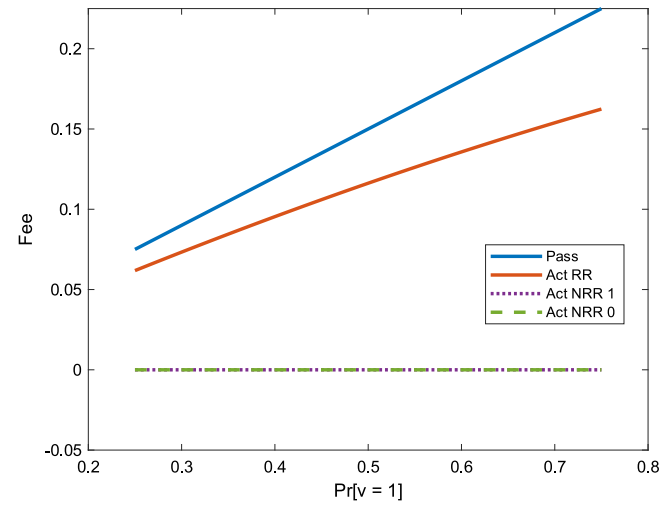

Fig. 3. Equilibrium Fees.
Fig. 3 plots the equilibrium fee f^* against the probability that $v = 1$ (i.e., ρ). The blue line represents the passive lender ("Pass"), the solid orange line represents the active lender with recall risk ("Act RR"), the dotted purple line represents the active lender with no recall risk with asset demand $D_L^*(b = 0) = 1$ ("Act NRR 1"), and the dashed green line represents the active lender with no recall risk with asset demand $D_L^*(b = 0) = 0$ ("Act NRR 0"). I also set $\bar{C} = \frac{3\rho}{5}$.

the lender's expected profits, so he wants to move as little as possible from that amount while still satisfying (11). The lender will prefer this choice to a fee that violates (11) because the latter is equivalent to setting an infinitely high fee in the sense that both eliminate short selling. On the other hand, when the lender sells following no loan agreement, his optimal fee without recall risk already satisfies (11) because he intentionally sets a high fee to reduce trading competition from the speculator.

I now examine how the active lender's need to commit impacts the equilibrium.[27]

Proposition 4. *The unique equilibrium trading strategy for the active lender with recall risk is to sell conditional on a loan agreement and not to trade otherwise. Equilibrium fees equal $\Delta E[P(z)]$.*

Besides the higher fee, the main difference between Propositions 3 and 4 is that buying conditional on no loan agreement ($D_L(b = 0) = 1$) is no longer an equilibrium trade for the active lender. This change is a consequence of Lemma 2, which states that the lender's valuation of the asset conditional on no loan agreement is decreasing in the fee. Thus, when the lender is forced to raise his fee from $f^* = 0$ to $f^* = \Delta E[P(z)]$ as a commitment device, buying becomes less profitable.

The following corollary then follows from Lemma 3 and Proposition 4.

Corollary 2. *The active lender's expected profits strictly decrease when recall risk is introduced.*

Essentially, giving the lender the ability to recall shares harms his expected profits, as it forces him to deviate from his profit-maximizing fee with no offsetting benefit. As a result, the lender would prefer to write a contract waiving his right to recall shares—if allowed.

3.5. Comparison of equilibria

There are three cases to examine: (1) the passive lender, (2) the active lender without recall risk, and (3) the active lender with recall risk.

Fig. 3 plots equilibrium fees: the main takeaway is that the passive lender always charges higher fees than the active lender. Without recall risk, the passive lender chooses the revenue-maximizing fee. The active lender, by contrast, lowers fees to zero because doing so enables him to extract more information from the lending market. However, when lending agreements allow recall (as they must for U.S. mutual funds in practice), this lower fee also makes it less costly for the active lender to recall the share, so he must raise it to reestablish commitment. The passive lender's fee is unaffected by recall risk because he can inherently commit not to recall shares. Nevertheless, the increased active fee is still lower than the unchanged passive fee because the two lenders have different fee-setting objectives: the passive lender aims to maximize lending revenue, while the active lender is willing to sacrifice some of that revenue for better information. As a result, the active lender optimally chooses the minimum fee enabling commitment ($f^* = \Delta E[P(z)]$), which is still below the revenue-maximizing level.

We see the equilibrium likelihood of the speculator not selling when $v = 0$ (i.e., $1 - G(c^*)$) in Fig. 4. As a benchmark, recall that the speculator would always sell when $v = 0$ if endowed with a share given the definition of $\bar{C}$ (see Assumption 2). Consequently, $1 - G(c^*)$ also serves as a measure of equilibrium short-sale constraints: higher values indicate tighter constraints with a value of zero corresponding to frictionless shorting.

The active lender imposes two costs on the short seller: (1) the fee and (2) a reduced expected selling price due to information leakage. By contrast, the only cost associated with the passive lender is the fee. Without recall risk, the active lender sets the fee equal to zero. This low fee compensates the short seller for the reduced selling price, allowing the active lender to attract more short sales when he does not trade following no loan agreement. When the active lender buys following no loan agreement, the zero fee is insufficient compensation. This result follows because the short seller's expected selling price $E[P(z)|D_S + D_L = -2] = \frac{1}{5}\sum_{z=-4}^{0} P(z)$ is lower when the lender buys following no loan agreement than when he does not trade (see Corollary 1). When recall risk is present, the active lender must raise fees as a commitment device, which limits his ability to compensate the

[27] The passive equilibrium does not change with recall risk.

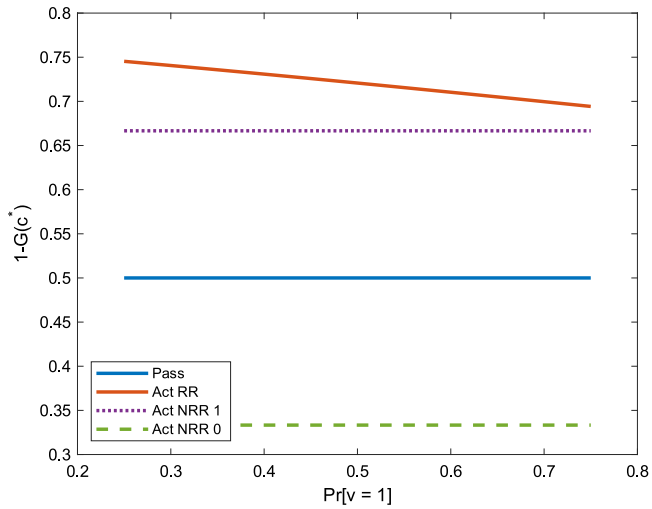

Fig. 4. Equilibrium Percentage Reduction in Short Selling.
Fig. 4 plots the equilibrium reduction in short selling $1 - G(c^*)$ against the probability that $\upsilon = 1$ (i.e., ρ). The blue line represents the passive lender ("Pass"), the solid orange line represents the active lender with recall risk ("Act RR"), the dotted purple line represents the active lender with no recall risk with asset demand $D_L^*(b = 0) = 1$ ("Act NRR 1"), and the dashed green line represents the active lender with no recall risk with asset demand $D_L^*(b = 0) = 0$ ("Act NRR 0"). I also set $\bar{C} = \frac{3\rho}{5}$.

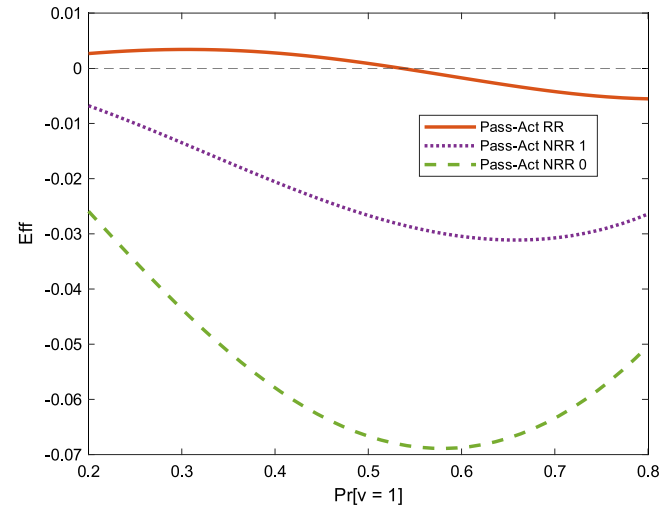

Fig. 5. Equilibrium Price Efficiency.
Fig. 5 plots the difference in price efficiency when the lender is passive vs. active. The solid orange line represents the difference in Eff between the passive lender and the active lender with recall risk ("Pass-Act RR"), the dotted purple line represents the difference in Eff between the passive lender and the active lender with no recall risk and active asset demand $D_L^*(b = 0) = 1$ ("Pass-Act NRR 1"), and the dashed green line represents the difference in Eff between the passive lender and the active lender with no recall risk and active asset demand $D_L^*(b = 0) = 0$ ("Pass-Act NRR 0"). I also set $\bar{C} = \frac{3\rho}{5}$.

short seller for the information leakage. As a result, the passive lender attracts greater shorting demand.

I formalize these insights in the following proposition.

Proposition 5. *The passive lender charges a higher fee than the active lender. Nevertheless, the passive lender also attracts more short selling than the active lender when recall risk is present. Without recall risk, the passive lender only attracts more short selling when the active lender buys following no loan agreement ($D_L^*(b = 0) = 1$).*

Proposition 5 helps reconcile a set of puzzling empirical findings: (1) active lenders trade on lending market information (Honkanen, 2025 and Greppmair et al., 2024); (2) greater passive ownership of a stock is associated with higher lending fees (Sikorskaya, 2025, Palia and Sokolinski, 2024, and von Beschwitz et al., 2025); and (3), short-selling demand increases with passive ownership (von Beschwitz et al., 2025, Sikorskaya, 2025, and Palia and Sokolinski, 2024). These findings suggest that active lenders' trades are more costly to short sellers than the higher fees of passive lenders. However, the economic forces that prevent active lenders from sufficiently lowering fees to attract greater shorting demand remain unclear. Active lenders clearly value lending market information, so what prevents them from lowering fees as much as possible to maximize learning? Proposition 5 provides an explanation: active lenders must keep fees above a certain level to credibly commit not to recall shares.

Finally, I look at price efficiency, which I define as

$$Eff \equiv 1 - E[|\upsilon - E[P(z)|\upsilon]|] \tag{12}$$

The term $|\upsilon - E[P(z)|\upsilon]|$ measures the distance between the expected price conditional on the value and the value. I subtract this expression from one so that my measure has the property that higher values indicate greater efficiency. As a benchmark, a perfectly revealing price has an efficiency measure of one.

Proposition 6. *Without recall risk, price efficiency is always higher under the active lender. With recall risk, price efficiency can be higher under the passive lender whenever ρ is sufficiently low.*

Proposition 6 states that price efficiency can be higher under a passive lender when recall risk is present. The active lender's incentive to recall shares forces him to raise fees as a commitment device. These higher fees, combined with the fact that the active lender front-runs short sales, exacerbate short-sale constraints to such an extent that the passive lender — who never trades — can actually improve price efficiency. More specifically, the passive lender improves price efficiency when the ex-ante probability that $\upsilon = 0$ is high (see Fig. 5). This result arises because relaxed short-sale constraints have a greater effect on price efficiency when short sales are more likely (i.e., it is more likely that $\upsilon = 0$).

However, even when passive lenders do not improve price efficiency, the broader takeaway from this analysis is that their participation in the securities lending market helps mitigate the harm caused by their lack of informed trading. As a result, models examining how the rise of indexing affects price efficiency overstate its negative impact if they ignore the securities lending market by, for example, assuming frictionless short selling.

4. Extensions

4.1. Lender reputation

It is perhaps unsurprising that lenders behave so opportunistically in my main model. There is no future relationship or reputation to maintain, so active lenders exploit lending market information to improve short-term trading profits. As a result, fees must adjust, sometimes dramatically, to account for this behavior. It is then natural to ask whether active lenders would behave the same way in an infinitely repeated game, where they consider not only current-period profits, but also future profits. Can a reduced continuation value for lenders, upon detection of front-running or strategic recall, deter their myopic behavior?

Front running is unlikely to be deterred for two reasons. First, if the short seller's ex-post information set contains only her trading price, perfect detection of front running is impossible. For example, the

speculator expects an order flow between 1 and −3 when she shorts and the lender never trades. If the lender deviates and sells after lending, the resulting order flow will be between 0 and −4. The only way the speculator can detect this deviation is if the price outcome is $P(-4)$ *and* $P(-4) \neq P(z)$ for $z \in \{1, 0, \dots, -3\}$. However, with "unprejudiced" beliefs about off-equilibrium order flows (as discussed in Section 2.3), $P(-4)$ will always equal $P(-3)$, meaning that the speculator cannot detect this deviation.[28]

It is also not obvious that the short seller would prefer her lender to refrain from trading on lending market information, since doing so would result in higher fees. If the active lender does not trade, he effectively becomes a passive lender and sets fees according to Eq. (5). As shown in Fig. 3, passive fees always exceed active fees, and in the absence of recall risk, short-selling profits can be worse with a passive lender (see Proposition 5). One can, therefore, view the active lender as buying the short seller's information with a reduced fee.

Unlike front running, strategic recall is both observable by and unambiguously detrimental to the short seller. The recall decision is communicated directly from the lender to the borrower, leaving no uncertainty, and recall always harms the short seller's profits. Applying standard results from infinitely repeated games, the lender will refrain from recalling shares in equilibrium if and only if the following reduced-form condition is satisfied:

$$f + E[P(z)|D_S + D_L = -2] + \frac{\delta}{1-\delta}\overline{\pi} > E[P(z)|D_S + D_L = -1] + \frac{\delta}{1-\delta}\underline{\pi}$$
$$\implies \quad f > \Delta E[P(z)] - \frac{\delta}{1-\delta}(\overline{\pi} - \underline{\pi}) \qquad (13)$$

Here, δ is the lender's discount factor, $\overline{\pi}$ is the lender's expected per-period profit when he has never recalled shares, and $\underline{\pi}$ is the lender's expected per-period profit when he has previously recalled shares. The short seller plays a "trigger" strategy: she behaves as if there is no recall risk if the lender has never recalled shares before, but reverts to only accepting fees that satisfy the static no-recall condition given by Eq. (11) if the lender has previously recalled shares. Thus, $\overline{\pi} \geq \underline{\pi}$ because the lender can set the fee closer to his profit-maximizing level (as described by (10)) if he has never recalled shares. If the lender cares sufficiently about the future (i.e., has a large δ), the threat of a permanent switch to the "bad" recall-proof equilibrium from Proposition 4 is enough to deter recall—even when fees are low. I discuss which factors are likely to impact δ empirically in Section 5.

4.2. Lender selling recalled shares

In my base model, I do not allow the active lender to sell recalled shares. However, let us now relax that assumption and see how being able to sell recalled shares affects his recall decision.

As an illustrative example, suppose the lender is endowed with N shares. Further, suppose a candidate equilibrium includes the speculator shorting M shares when $v = 0$ and the lender selling $N - M - \omega$ out of his remaining inventory after making the loan, where ω is an integer between zero and $N - M - 1$.[29] The speculator's expected profits following the loan agreement are $M(E[P(z)|D_S + D_L = -N + \omega] - f)$, and the lender's are $(N - M - \omega)E[P(z)|D_S + D_L = -N + \omega] + Mf$. Observe that the lender always has the option to recall and sell all M shares to earn an extra $ME[P(z)|D_S + D_L = -N + \omega]$ (where this time $D_S = 0$ and $D_L = -N + \omega$) in trading profits in exchange for losing Mf in lending revenue. Thus, the lender requires $f \geq E[P(z)|D_S + D_L = -N + \omega]$ to prefer not recalling the shares, which is precisely the condition that eliminates all short selling. Even if selling all M recalled shares is not the lender's optimal strategy, it remains a feasible option.[30] If a more profitable recall and trading strategy exists, the lender would then require even higher fees to commit.

It is worth noting that the complete unraveling of the lending market emerges because the lender and short seller value the asset equally (i.e., at zero) after the loan agreement. When they have the same valuation, the lender can expropriate all of the short seller's expected trading profits by recalling shares in exchange for the fee revenue. Thus, commitment requires fee revenue greater than the short seller's expected trading profits, which necessarily eliminates shorting.

If the lender's valuation remains above that of the short seller (e.g., by adding some noise to the lending market signal), the commitment problem is less extreme but still substantial. Suppose the lender's post-loan valuation equals some amount $\underline{v_L}$ slightly above zero. In this case, repeating the same steps as before shows that commitment requires fees at least as high as $E[P(z) \mid D_S + D_L = -N + \omega] - \underline{v_L}$, so the short seller retains, at best, only the portion of her profits that comes from her information advantage over the lender. If the lender has an even more profitable strategy than selling all M recalled shares, the required fee rises further—potentially to the point of destroying the lending market again.

In short, allowing the active lender to sell recalled shares exacerbates the commitment problem examined in Section 3.4 to such an extent that short selling no longer occurs in equilibrium. This striking outcome raises a natural question: How can active lenders lend shares in practice despite this problem?

One answer may lie in the institutional details of the securities lending market: selling recalled shares before regaining possession exposes the lender to settlement risk, as delivery depends on the borrower returning them on time. If this risk is sufficiently high, the lender may refrain from selling recalled shares. Second, the lender's reputational concerns (discussed in Section 4.1) can also help mitigate the commitment problem, as a reduced continuation value following recall reduces the fee required to sustain commitment. These repeated-game considerations become especially important when the lender can sell recalled shares.

Third, portfolio constraints also facilitate commitment. Buffa et al. (2022) provide strong evidence that active funds are often limited in how far they can deviate from a benchmark index. Microfoundations for such constraints include fund managers responding optimally to performance contracts benchmarked to an index (Kashyap et al., 2023) and uncertainty about fund manager skill (Buffa et al., 2022). An unintended benefit of such constraints is that they prevent the fund manager from exploiting lending market information too aggressively. By limiting the number of shares the active manager can sell after learning bad news, portfolio constraints enable him to commit not to recall shares at lower fees, thereby supporting greater equilibrium short selling.

In sum, while Sections 4.1 and 4.2 reinforce the insight that higher fees ensure a commitment not to recall shares, whether such fees also support equilibrium short selling depends on other features of the lender's environment (e.g., his ability to sell recalled shares, reputation concerns, portfolio constraints, market structure, etc.). Put differently, the fee is the instrument that determines whether a commitment not to recall shares is incentive compatible, but the precise level at which this commitment is achieved — and whether that level still supports short selling — depends on the lender's environment.

[28] Even if we expand the speculator's ex-post information set to include the realization of z, there is still only a $\frac{1}{5}$ probability of detection. The detection probability is almost certainly lower in actual financial markets due to the presence of many more traders who can alter the resulting order flow.

[29] The active lender will always sell at least one share due to Lemma 1.

[30] The tradeoff between price impact and trading quantity is a standard problem for informed traders in Kyle (1985) style microstructure models. Thus, it is not obvious that selling all M recalled shares is the lender's optimal decision.

4.3. Informed active lender

The base model assumes that the active lender is uninformed, but let us now relax that assumption and instead allow him to observe some private signal $y \in \{0, 1\}$ about the payoff v where $Pr[y = v|v] = q \in [0.5, 1]$. My base model is therefore the case where $q = 0.5$. Notation-wise, I denote L_y as the lender who sees signal y, $D_L^y(b)$ as L_y's asset demand conditional on lending market signal b, and c_y^* as the shorting cutoff for the short seller borrowing from lender L_y.[31]

The core insights from this extension are clearest in the limiting case where the lender's private signal approaches perfect accuracy (i.e., as q approaches one).[32] I discuss the imperfect private signal case in Online Appendix E. Naturally, as the lender is nearly perfectly informed, he will buy following no loan agreement when his private signal is positive ($D_L^{1*}(b = 0) = 1$) and sell when negative ($D_L^{0*}(b = 0) = -1$). The lender continues to sell following a loan agreement because the speculator is still perfectly informed. The following lemma describes how the lender's private signal impacts his fee choice.

Lemma 4. *Without recall risk, the informed active lender's optimal fee remains as described by Eq. (10). With recall risk, the informed active lender's optimal fee remains as described by Lemma 3.*

Lemma 4 essentially states that the lender's private signal has no direct effect on his optimal fee as a function of prices.[33] This result follows because the fee only affects the lender's expected profits conditional on $v = 0$. As a result, the private signal — which influences the lender's prior (i.e., pre-lending market) belief that $v = 0$ — is a multiplicative constant in his first-order condition, which cancels out. This lemma then implies the following corollary.

Corollary 3. *The negatively informed lender (L_0) always sets a fee at least as high as the positively informed lender (L_1) and sets a strictly higher fee whenever L_0 attracts a positive amount of short sellers in equilibrium (i.e., when $c_0^* > 0$).*

Lemma 4 and Eq. (10) tell us that the positively informed L_1 chooses $f_1^* = 0$ while the negatively informed L_0 wants $f_0^* = \frac{E[P(z)|D_S+D_L=1]}{2} > 0$ when there is no recall risk. With recall risk, Lemmas 3 and 4 imply that the positively informed L_1 now chooses $f_1^* = \Delta E[P(z)]$ since he must commit not to recall shares; the negatively informed L_0 does not need to change his fee because it is already sufficiently high. Thus, $f_0^* = \frac{E[P(z)|D_S+D_L=1]}{2} \geq \Delta E[P(z)] = f_1^*$ when lenders can recall shares. Additionally, $f_0^* > f_1^*$ whenever the negatively informed L_0 attracts short sellers in equilibrium (i.e., $c_0^* > 0$) because

$$
\begin{aligned}
& c_0^* > 0 \\
\iff & E[P(z)|D_S + D_L = -2] - \frac{E[P(z)|D_S + D_L = 1]}{2} > 0 \\
\iff & \frac{E[P(z)|D_S + D_L = 1]}{2} > E[P(z)|D_S + D_L = -1] \\
& \quad - E[P(z)|D_S + D_L = -2] \equiv \Delta E[P(z)] \\
\iff & f_0^* > f_1^*
\end{aligned}
$$

Corollary 3 is also consistent with the empirical evidence that high lending fees predict low subsequent returns (e.g., Cohen et al., 2007 and Jones and Lamont, 2002). Intuitively, when $v = 0$, the lender is more likely to receive a negative private signal and set a high fee. While the empirical literature typically attributes the negative relationship between fees and returns to greater borrowing demand when stocks are overvalued, my model shows that the lender's decisions also play a crucial role. In this respect, Corollary 3 aligns with Duong et al. (2017), who find that lenders preemptively set high fees in anticipation of elevated borrowing demand and low subsequent returns.

Finally, I document in Online Appendix E that my results about passive lenders potentially improving price efficiency do not depend on the active lender being uninformed. In general, there is a cutoff level of information quality for the active lender below which passive lenders improve price efficiency.[34]

4.4. Lender competition

I assume that the lender is a monopolist who can make a take-it-or-leave-it offer to the speculator in the base model. I now examine how my results change when lenders engage in Bertrand fee-setting competition with each other. Suppose two lenders simultaneously offer fees to the speculator at stage 3a of my timeline from Section 2.2. At stage 3b, the speculator either borrows a single share from one of the lenders or borrows from neither of them. Each lender observes only whether the speculator borrowed from him; that is, a lender cannot distinguish between losing the competition to the other lender and the speculator not borrowing at all.

I consider three cases: (1) competition between active lenders, (2) competition between an active and a passive lender, and (3) competition between passive lenders.

First, I consider competition between active lenders. Strikingly, this competition has no impact on fees—both with and without recall risk. Consider first the case without recall risk. From Proposition 3, the active lender already sets fees equal to marginal cost (i.e., zero) as a monopolist to acquire more precise information about the stock. Thus, competition will have no effect on fees since they cannot drop any lower. The logic is similar with recall risk: competition cannot lower fees because each lender is already selecting their lowest possible fee $f^* = \Delta E[P(z)]$. If an active lender tried to undercut this fee, the short seller would refuse to borrow from him, anticipating that he would recall the loan (see Fig. 2). Thus, competition cannot bring active lenders' fees below the no-recall lower bound of $\Delta E[P(z)]$.

Now suppose an active and a passive lender are competing against each other. As in the previous case, the active lender is already choosing his lowest possible fee, so the key question is how competition affects the passive lender. With their monopolistic fee choices, the active lender can be more attractive to the short seller when there is no recall risk (see Proposition 5).[35] This outcome occurs because the active lender's optimal fee is zero while the passive lender chooses the revenue-maximizing fee of $\frac{E[P(z)|D_S+D_L=-1]}{2}$. Thus, the passive lender will want to lower his fee by just enough to win the competition. The active lender then cannot respond in turn because he is already setting his fee equal to zero.

The essential insight from this case is that the passive lender does not need to undercut the active lender's fee to win the competition. The short seller is concerned about the *total* cost of borrowing, which includes the fee and the cost of information leakage. Because passive lenders do not trade, they impose no information leakage costs on the speculator. As a result, the passive lender does not need to charge a lower fee to win, but only a fee low enough to ensure that he imposes a lower total cost.[36] This competitive advantage enables the passive lender to charge a higher fee and still win the speculator's business. The

[31] Observe that the speculator does not care about the active lender's signal realization per se because she is perfectly informed about v and both L_1 and L_0 trade the same way after making a loan. Consequently, only the observable fee matters for the speculator's shorting decision.

[32] I focus on the limiting case where q approaches one rather than $q = 1$ because when $q = 1$, L_1 knows there will be no short-selling demand and thus has an undefined optimal fee.

[33] The signal does, however, have an indirect effect on fees via the equilibrium prices.

[34] See Proposition E.3 in Online Appendix E for more details.

[35] This result occurs when the active lender demands $D_L^*(b = 0) = 0$.

[36] For example, suppose the active lender sells following a loan agreement and does not trade otherwise. Then expected shorting profits with the active

same logic applies when recall risk is present. In this case, the active lender cannot reduce his fee below $\Delta E[P(z)]$, so the passive lender will need to reduce his fee by less to remain competitive. In fact, Proposition 5 implies that the passive lender would win the competition even with his monopolistic fee choice.

Finally, consider competition between two passive lenders. This case yields the textbook Bertrand outcome: competition drives fees down to marginal cost, which in this setting is zero. This result implies that efforts to increase competition in the lending market will have the greatest impact on fees when the lenders involved are passive, as they are the only ones using their market power to charge elevated fees.

5. Discussion

5.1. Empirical predictions

My model makes predictions about how lending market information affects lenders' trades. First, active securities lenders sell stocks they have on loan, consistent with Honkanen (2025) and Greppmair et al. (2024). Second, my model generates new predictions about how fees relate to lenders' trades on stocks made available for loan but not taken up. Eq. (10) states that high fees and low utilization predict lender sales, while low fees and low utilization predict lender purchases. Intuitively, lenders who plan to sell raise fees to limit trading competition, whereas lenders who plan to buy prefer low fees to make better informed purchases. This result also generates the well-known relationship between high lending fees and low returns (e.g., Cohen et al., 2007 and Jones and Lamont, 2002) because the only time a lender sells without a loan agreement in equilibrium is when he has a strong negative signal (see Section 4.3). However, in my model, high fees emerge in anticipation of shorting demand rather than in response to it, in line with Duong et al. (2017).

My model also predicts that strategic recall risk increases lending fees. Active lenders are tempted to recall shares to improve their selling price and must raise fees to commit not to. Passive lenders have no such incentive because they cannot trade. Thus, since lenders cannot contractually commit not to recall shares, they must do so endogenously through higher fees.

I expect my results to be more salient for lenders with an in-house lending agent. I have abstracted from the lending agent in this paper; however, in practice, shareholders deposit shares they want to lend with a lending agent, who is then responsible for lending them out. This lending agent is often a third-party dealer bank. Funds may choose to use a third party because establishing a lending program can be expensive. However, large mutual fund families — such as Fidelity, BlackRock, and Vanguard — manage their own lending operations. I would therefore expect in-house lenders to show greater care in tailoring fees to maximize fund (or fund family) profits as described in this paper, compared to third-party agents. This insight also suggests that switching from a third-party to an in-house lending agent is about more than just costs: there are also informational benefits from optimal fee setting. To what extent in-house and third-party lending agents differ in their fee-setting approaches is an interesting question for future research.

As mentioned in Section 4.1, the lender's time preference parameter δ from Eq. (13) affects his willingness to recall shares in a repeated game. Specifically, he will not recall the share when $f > \Delta E[P(z)] - \frac{\delta}{1-\delta}(\overline{\pi} - \underline{\pi})$. The time-preference parameter δ has many empirical interpretations. First, one can view the discount factor δ as the probability that the game will continue for another period. Suppose there are two types of speculators: a high type who will continue playing next period with probability $\delta^h > 0$ and a low type who will continue with probability zero. One can view the high types as the lender's long-term customers, while the low types are one-off short sellers. Naturally, the lender will be less inclined to recall a high type's share because he expects future business from her. Thus, from (13), we expect repeat customers to get lower fees. Suppose further that the speculator's type is only known by her current lender. High-type speculators would be hesitant to switch lenders because a new lender would not know their type and would therefore require a greater fee not to recall the loan. The implication that the incumbent is the only lender who can commit not to recall loans at low fees is a novel source of market power in the short-selling literature.

Agency concerns may also influence the lender's discount factor from Section 4.1. Suppose the lender is a mutual fund. The fund manager cares about his fund's performance to the extent that it impacts his present and future compensation. As a result, the fund manager's optimal recall strategy (i.e., his discount factor δ) may differ from that of his investors. For example, two recent empirical papers show that fund manager pay is a concave function of fund revenue (Ibert et al., 2017 and Cen et al., 2024). Such a relationship suggests that a fund manager will be more likely to act opportunistically and recall shares when current performance is weak.

While an index fund manager cannot trade on information gained from the lending market, he should be able to pass it along to an active fund in the same family. There is mixed empirical evidence for this conjecture: (Honkanen, 2025) finds evidence for these within-family spillovers in the U.S., while Greppmair et al. (2024) find none in Germany. My model predicts that index funds that share lending market information with others will charge earn lower fee revenue in equilibrium because short sellers will anticipate the information leakage and require lower fees to borrow.

Finally, consider the case where the short seller has to close out their position before the fundamental value v is revealed. It seems likely that this close-out price will be higher with a passive lender because he does not trade.[37] Thus, the very feature that makes passive lenders attractive when opening a short position (higher prices) also makes them unattractive when closing out. However, short sellers will not stand idly by and hope that their (correct) information gets impounded into prices; instead, activist short sellers often proactively publicize their positions and release research reports explaining their reasoning.[38] Assuming such disclosure is costly, short sellers should be more willing to bear those costs when the no-disclosure price is expected to be higher, which, as previously discussed, should occur under passive lenders.[39] Thus, we should expect activist short sale disclosures to become more frequent as index investing rises.

5.2. Policy implications

Throughout the paper, I exploit the institutional detail that lenders governed by the 1940 Investment Company Act (e.g., U.S. mutual funds) must retain the ability to recall lent-out shares at will. This regulation is driven in part by corporate governance concerns, as lenders cannot cast votes associated with lent-out shares. Thus, the regulation enables lenders to recall shares before important corporate votes to fulfill their stewardship responsibilities. However, my analysis reveals that this flexibility comes at a cost: since lenders cannot contractually commit not to recall shares, they must instead increase fees as a

lender are $E[P(z)|D_S + D_L = -2]$ when $f = 0$, so the passive lender only needs to set a fee below $E[P(z)|D_S + D_L = -1] - E[P(z)|D_S + D_L = -2]$ to win.

[37] See Online Appendix D for a microfoundation of this case.

[38] See Ljungqvist and Qian (2016) evidence of short sellers releasing research reports to convince existing shareholders to sell.

[39] These costs can include the direct costs of compiling a credible report and the indirect legal or reputational costs associated with publicly declaring that a firm is overvalued. See Lamont (2012) for more details.

commitment device. As a result, this regulation inadvertently raises the cost of borrowing shares.

Nevertheless, a policymaker might view increased lending fees — and the associated decline in price efficiency — as an acceptable cost if it leads to more effective stewardship by lenders. Suppose the lender can create some governance benefit by recalling and voting the lent-out share. The short seller would anticipate this additional incentive to recall the share and would not borrow unless the fee is high enough to deter the lender from doing so. Thus, recall still never occurs in equilibrium because the fee would increase until it fully compensates the lender for the forgone governance benefit (and trading gains, if active). Importantly, this instance of recall risk also applies to passive lenders: while they do not trade on lending market information, they can still recall shares to vote. While a full welfare analysis of this regulation is beyond the scope of this paper, my framework highlights a significant downside of this contractual inflexibility.[40]

Lender market power is also a growing concern in the securities lending market since it can result in higher lending fees (e.g., Chen et al., 2025). The rise of index investing may be partly responsible for this trend, as flows into index products often end up at large fund families, such as the "big three" of Vanguard, BlackRock, and State Street. A natural consequence of this development is that these fund families also play a larger role in the lending market. Section 4.4 analyzes how competition between lenders affects equilibrium fees and shows that efforts to promote competition are especially effective when that competition is between passive lenders. This result arises because only passive lenders exploit their market power to charge high fees; active lenders instead prefer low fees to improve the quality of their information. Consequently, my analysis suggests that the rise of passive investing makes efforts to improve competition in the lending market more effective for two reasons. First, index investing may contribute to greater lending market concentration. Second, even holding the level of concentration fixed, heightened competition has a larger effect on fees when lenders are passive.

6. Conclusion

In this paper, I model securities lending as an information acquisition problem, where lenders infer information about a stock from the level of borrowing demand. Active lenders, who maximize the sum of lending revenue and trading profits, can extract more information by reducing fees from the revenue-maximizing level. However, low fees also make it less costly for them to recall shares to disguise their trades. As a result, active lenders must maintain fees above a certain level as a commitment device to continue to attract short sellers. The combined costs of the fee and information leakage harm short sellers so much that passive lenders, who do not trade, can relax short-sale constraints and potentially improve price efficiency—despite charging higher fees.

Information leakage in the securities lending market is a relatively unexplored research topic, perhaps because securities lending occurs in an opaque, over-the-counter setting, making it difficult to find quality data. However, the SEC has recently taken steps to increase transparency. Rule 10c-1a (adopted in October 2023) requires lenders to report loan-level quantities and fees for the first time.[41] Assuming these rules survive ongoing legal challenges, the availability of such granular data should spur further research in this area. My model provides a tractable theoretical framework for motivating and interpreting such research and demonstrates how a fund's lending and trading activities are connected.

[40] Moreover, it is also not obvious that recalling shares makes lenders effective stewards because other, more informed market participants may better utilize these votes. E.g., Christoffersen et al. (2007).

[41] https://www.sec.gov/files/34-98737-fact-sheet.pdf

CRediT authorship contribution statement

George C. Nurisso: Writing – review & editing, Writing – original draft, Visualization, Methodology, Investigation, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Proofs from main text

Supplementary results

Result S1: *Price Calculations*

Since virtually every proof requires calculating prices, I will describe how to do so here in full detail. I will not repeat these steps in the proofs to follow to avoid excessive repetition; one can calculate prices there by plugging the conjectured trading strategies into this generic procedure. Without loss of generality, I will calculate prices with an active lender; the passive lender is just a special case where D_L is always zero.

From the main text, the price that order flow z maps to is

$$P(z) = Pr[v = 1|z] = \frac{Pr[z|v = 1]Pr[v = 1]}{Pr[z|v = 1]Pr[v = 1] + Pr[z|v = 0]Pr[v = 0]} \tag{14}$$

Observe that $Pr[v = 1] = \rho$ and $Pr[v = 0] = 1 - \rho$. We next need to calculate the two $Pr[z|v]$ terms. Expanding gives us:

$$Pr[z|v = 0] = G(c^*)Pr[z|v = 0, c < c^*] + \left(1 - G(c^*)\right) Pr[z|v = 0, c \geq c^*] \tag{15}$$

where c^* is the speculator's shorting cutoff. Recall that the speculator always buys when $v = 1$ due to Assumption 1, so there is no analogous breakdown of $Pr[z|v = 1]$.

It is at this point that the conjectured trading strategies come into play. For $Pr[z|v = 0, c < c^*]$, we know that $D_S = -1$ given the speculator's threshold trading strategy and that the lender's demand will equal $D_L^*(b = 1)$. Denote $w^{-1} \equiv -1 + D_L^*(b = 1)$. Then

$$Pr[z|v = 0, c < c^*] = \begin{cases} \frac{1}{5} \text{ if } z \in \{w^{-1}, w^{-1} \pm 1, w^{-1} \pm 2\} \\ 0 \text{ otherwise} \end{cases} \tag{16}$$

The $\frac{1}{5}$ probabilities come from the possible realizations of aggregate noise demand. Next we have the case where the speculator does not trade (i.e., when $v = 0$ and $c \geq c^*$). Since there is no loan agreement, the lender's demand will equal $D_L^*(b = 0)$. Denote $w^0 \equiv 0 + D_L^*(b = 0)$ as the sum of the lender's and the speculator's demands in this case. Then

$$Pr[z|v = 0, c \geq c^*] = \begin{cases} \frac{1}{5} \text{ if } z \in \{w^0, w^0 \pm 1, w^0 \pm 2\} \\ 0 \text{ otherwise} \end{cases} \tag{17}$$

Lastly, we have the final case where the speculator buys. Since there is no short sale, the lender's demand will still equal $D_L^*(b = 0)$. Denote $w^1 \equiv 1 + D_L^*(b = 0)$ as the sum of the lender's and the speculator's demands in this case. We then have

$$Pr[z|v = 1] = \begin{cases} \frac{1}{5} \text{ if } z \in \{w^1, w^1 \pm 1, w^1 \pm 2\} \\ 0 \text{ otherwise} \end{cases} \tag{18}$$

We can now plug Eqs. (16) and (17) into (15), and (15) and (18) into (14) to get our price $P(z)$. We can repeat this procedure for all possible

realizations of z to get our entire set of prices. Expected prices are then calculated by averaging over these $P(z)$'s. For example,

$$E[P(z)|D_S + D_L] = \frac{1}{5}\sum_{z=D_S+D_L-2}^{D_S+D_L+2} P(z) \tag{19}$$

Proposition S1 *Suppose the speculator is endowed with a share, so she never needs to borrow. Then the speculator will always sell when $v = 0$ if and only if $\bar{C} \leq \frac{3\rho}{5}$. If this condition is met, then the unique equilibrium in this setting is for the speculator to always trade in the direction of her signal and for the lender to not trade. Assumption 2 therefore implies $\bar{C} = \frac{3\rho}{5}$.*

Proof. Recall that by Assumption 1, the speculator always buys when $v = 1$. I first calculate the maximum level of c that supports the speculator always selling when $v = 0$ in this setting. Given these conjectured trading strategies, I can calculate prices by following the procedure described in Result S1[42]:

$$P(z) = \begin{cases} 1 \text{ when } z \in \{2,3,4\} \\ \rho \text{ when } z \in \{-1,0,1\} \\ 0 \text{ when } z \in \{-4,-3,-2\} \end{cases}$$

The speculator's expected selling price when $v = 0$ is $E[P(z)|D_S + D_L = -1] = \frac{3\rho}{5}$. Thus, the speculator will always sell when $v = 0$ if $\bar{C} \leq \frac{3\rho}{5}$.

It is then straightforward to show that $D_L^* = 0$ (note that demand is no longer conditional on b without the lending market) is always an equilibrium trading strategy for the lender. The lender's expected buying profits equal: $\rho - \rho E[P(z)|D_S + D_L = 2] - (1-\rho)E[P(z)|D_S + D_L = 0] = \frac{\rho^2-1}{5} < 0$. Similarly, the lender's selling profits equal $\rho E[P(z)|D_S + D_L = 0] + (1-\rho)E[P(z)|D_S + D_L = -2] - \rho = \frac{\rho(\rho-2)}{5} < 0$. Thus, $D_L^* = 0$ is an equilibrium trade.

It then remains to show that there is no other equilibrium trading strategy for the lender. Suppose $D_L = 1$. Then, using the same process as before, we can calculate prices to find the lender's expected buying profits: these equal $\rho - \rho E[P(z)|D_S + D_L = 2] - (1-\rho)E[P(z)|D_S + D_L = 0] = 0$, so the lender never finds buying profitable.

The story is similar when we instead conjecture $D_L = -1$. We can calculate prices and see that the lender's expected selling profits equal $\rho E[P(z)|D_S + D_L = 0] + (1-\rho)E[P(z)|D_S + D_L = -2] - \rho = 0$, so the lender also never finds selling profitable.

Thus, the unique equilibrium in this counterfactual where the speculator is endowed with a share is for her to trade in the direction of her signal and for the lender to never trade. □

Proofs of Section 3

Proof of Lemma 1. See text. □

Proof of Proposition 1. That the passive lender's optimal fee maximizes expected lending revenue follows from his objective function (4). The speculator's shorting profits equal $E[P(z)|D_S + D_L = -1] - f = E[P(z)|D_S + D_L = -1] - \frac{E[P(z)|D_S+D_L=-1]}{2} = \frac{E[P(z)|D_S+D_L=-1]}{2} = \frac{3\rho}{10}$ from Eqs. (3) and (5) in the main text. Thus, the optimal fee takes half of the speculator's expected shorting profits. Again, from (3) and (5), c^* must also equal $\frac{E[P(z)|D_S+D_L=-1]}{2} = \frac{3\rho}{10}$. □

Proof of Lemma 2. From Eq. (8), $E[v|b=0] = \frac{\rho}{\rho+(1-\rho)(1-G(c(f)))}$ where $G(c(f)) = \frac{E[P(z)|D_S+D_L=-2]-f}{\bar{C}} \in [0,1]$. Note that the fee only appears in the valuation via $G(c(f))$. It then follows that since the valuation is increasing in $G(c(f))$ and $G(c(f))$ is decreasing in f, the valuation must be decreasing in f. □

[42] The prices $P(4)$ and $P(-4)$ follow from the unprejudiced beliefs refinement.

Proof of Proposition 2. Fees can be found by calculating the first-order conditions of Eq. (9) with respect to f. To do this, first note that $Pr[D_s = -1] = (1-\rho)G(c(f))$ with $G(c(f)) = \frac{E[P(z)|D_S+D_L=-2]-f}{\bar{C}}$. Then note that $E[v|b=0]$ is defined by Eq. (8). Finally, the expected prices conditional on $b = 0$ are:

$$E[P(z)|b=0] = \begin{cases} \frac{\rho E[P(z)|D_S+D_L=2]+(1-\rho)(1-G(c(f)))E[P(z)|D_S+D_L=1]}{\rho+(1-\rho)(1-G(c(f)))} \\ \quad \text{when } D_L(b=0) = 1 \\ \frac{\rho E[P(z)|D_S+D_L=0]+(1-\rho)(1-G(c(f)))E[P(z)|D_S+D_L=-1]}{\rho+(1-\rho)(1-G(c(f)))} \\ \quad \text{when } D_L(b=0) = -1 \end{cases}$$

The optimal fees given by (10) then immediately follow from the first-order condition of (9) with respect to f.

The only other thing I need to show is that when the lender's optimal fee violates the non-negativity constraint, his constrained-optimal fee is zero; this outcome only happens when $D_L(b=0) = 1$. The derivative of expected profits with respect to the fee is

$$\frac{Pr[v=0]\left(\frac{-E[P(z)|D_S+D_L=1]}{2} - f\right)}{\bar{C}} < 0$$

for all $E[P(z)|D_S + D_L = -2] - \bar{C} \leq f \leq E[P(z)|D_S + D_L = -2]$. Given how $\bar{C}$ is defined in Assumption 2, the inequality $E[P(z)|D_S + D_L = -2] - \bar{C} < 0$ will always hold. As a result, the lender's profits are strictly decreasing in f for all $f \geq 0$, so he will take the lowest possible fee that satisfies the non-negativity constraint: $f = 0$. □

Proof of Proposition 3. From Lemma 1, we know the lender's optimal trade conditional on $b = 1$ is to sell. Thus, we only need to discover the lender's equilibrium trades conditional on $b = 0$. There are three possible pure-strategy equilibrium trades for the lender conditional on $b = 0$: $D_L(b=0) \in \{-1,0,1\}$. Proposition 3 states that only the latter two can be part of the lender's equilibrium trading strategy.

First, I demonstrate that $D_L(b=0) = -1$ is not an equilibrium by showing that selling conditional on $b = 0$ is unprofitable for the lender. Given this proposed trading strategy, Result S1 gives us prices as a function of c^*.

$$P(z) = \begin{cases} 1 \text{ when } z = 2 \\ \frac{\rho}{\rho+(1-\rho)(1-G(c^*))} \text{ when } z = 1 \\ \rho \text{ when } z \in \{0,-1,-2\} \\ 0 \text{ when } z \in \{-3,-4\} \end{cases}$$

We pin down the cutoff c^* by plugging the lender's optimal fee (see Eq. (10)) into the indifference condition given by (6) from the main text and setting $c(f) = c^*$.

$$E[P(z)|D_S + D_L = -2] - \frac{E[P(z)|D_S + D_L = -1]}{2} = c^*$$

where expected prices are calculated by averaging over the relevant $P(z)$'s as defined in (19). The lender's expected selling profits conditional on $b = 0$ are:

$$E[P(z)|b=0] - E[v|b=0] = \frac{\rho E[P(z)|D_S + D_L = 0] + (1-\rho)(1-G(c^*))E[P(z)|D_S + D_L = -1] - \rho}{\rho + (1-\rho)(1-G(c^*))} \tag{20}$$

where $E[P(z)|D_S + D_L = 0] = \frac{1}{5}\left(1 + \frac{\rho}{\rho+(1-\rho)(1-G(c^*))} + 3\rho\right)$ and $E[P(z)|D_S + D_L = -1] = \frac{1}{5}\left(\frac{\rho}{\rho+(1-\rho)(1-G(c^*))} + 3\rho\right)$. Expanding (20) tells us that the lender's expected selling price is $\frac{2}{5}\left(\frac{\rho}{\rho+(1-\rho)(1-G(c^*))}\right) + \frac{3}{5}\rho$, which is always no greater than the lender's valuation of $\frac{\rho}{\rho+(1-\rho)(1-G(c^*))}$ (and strictly less than whenever $G(c^*) > 0$). Thus, it can never be an equilibrium for the lender to sell conditional on $b = 0$.

Table 1
Prices without recall.

	$D_L(b=0)=0$	$D_L(b=0)=1$
P(4)	1	1
P(3)	1	$\frac{\rho}{\rho+(1-\rho)(1-G(c^*))}$
P(2)	$\frac{\rho}{\rho+(1-\rho)(1-G(c^*))}$	$\frac{\rho}{\rho+(1-\rho)(1-G(c^*))}$
P(1)	$\frac{\rho}{\rho+(1-\rho)(1-G(c^*))}$	$\frac{\rho}{\rho+(1-\rho)(1-G(c^*))}$
P(0)	ρ	ρ
P(−1)	ρ	0
P(−2)	0	0
P(−3)	0	0
P(−4)	0	0

I now examine conditions under which the remaining two possible trading strategies are equilibria. First, I find prices for each proposed equilibrium by applying Result S1; I display them in Table 1. The two columns correspond to the conjectured equilibrium trading strategies. When an order flow is not reached in equilibrium, I apply the "unprejudiced beliefs" refinement to get the price (see Section 2.3).

I start with the $D_L(b=0)=0$ case. In this proposed equilibrium, the lender sets the fee equal to zero, which leads to the short-selling cutoff: $c^* = E[P(z)|D_S+D_L=-2] = \frac{2\rho}{5}$. With this cutoff, we can plug in to get equilibrium prices as a function of ρ and $\bar{C}$. I now verify that the conjectured trading strategy is optimal. Holding fees constant, the lender's profits from deviating to buying conditional on $b=0$ are:

$$E[v-P(z)|b=0] = \frac{\rho-\rho E[P(z)|D_S+D_L=2]-(1-\rho)(1-G(c^*))E[P(z)|D_S+D_L=1]}{\rho+(1-\rho)(1-G(c^*))} \tag{21}$$

where $E[P(z)|D_S+D_L=2] = \frac{1}{5}\left(2+2\frac{\rho}{\rho+(1-\rho)(1-G(c^*))}+\rho\right)$, $E[P(z)|D_S+D_L=1] = \frac{1}{5}\left(1+2\frac{\rho}{\rho+(1-\rho)(1-G(c^*))}+2\rho\right)$ and $G(c^*) = \frac{2\rho}{5\bar{C}}$. After some algebraic rearrangement, the lender will not deviate to buying so long as $\bar{C} \geq \frac{2\rho(1+2\rho)}{5(1+\rho)}$. This condition is always satisfied under Assumption 2.

Next, I show that the lender will never deviate to selling conditional on $b=0$. Observe from (20) that the deviation selling profits will be positive if and only if

$$\rho(E[P(z)|D_S+D_L=0]-1)+(1-\rho)(1-G(c(f)))(E[P(z)|D_S+D_L=-1]-0)>0 \tag{22}$$

where $E[P(z)|D_S+D_L=0] = \frac{1}{5}\left(2\frac{\rho}{\rho+(1-\rho)(1-G(c^*))}+2\rho\right)$ and $E[P(z)|D_S+D_L=-1] = \frac{1}{5}\left(\frac{\rho}{\rho+(1-\rho)(1-G(c^*))}+2\rho\right)$. When the lender deviates to selling conditional on $b=0$, he may also want to choose some new fee f' since the equilibrium fee may no longer be optimal. Let $G(c(f')) = \frac{E[P(z)|D_S+D_L=-2]-f'}{\bar{C}}$ be the resulting probability of a short sale when $v=0$ following this deviant fee f'. Observe that since the prices are fixed at their equilibrium value (since the market maker will not observe this deviation), (22) will be maximized when $G(c(f'))=0$. Thus, the expression on the left hand side of (22) can be no higher than $\rho E[P(Z)|D_S+D_L=-0]+(1-\rho)E[P(z)|D_S+D_L=-1]-\rho$. Plugging in, this expression equals $-\frac{3\rho^2}{10\rho+5}<0$, so deviating to selling is always unprofitable for the lender.

Next, I must verify that the lender will not deviate to a different fee and trading strategy. This verification procedure is simplified by Proposition 2, which states that there is a unique optimal fee for every trading strategy. As a result, I only need to check that the lender will not deviate to buying at the optimal buying fee or selling at the optimal selling fee (since he is already setting the optimal no-trade fee); any other fee and trading strategy pair is dominated by at least one of these three options. However, since the optimal buying fee is the same as the optimal no trade fee and my proof ruling out $D_L(b=0)=-1$ held for all possible deviant fee choices f', I have already ruled out these deviations.

Next, we move to the $D_L(b=0)=1$ case. The procedure to verify this equilibrium is the same as before. We can calculate the short-selling cutoff $c^* = E[P(z)|D_S+D_L=-2] = \frac{\rho}{5} < \bar{C}$ using the fact that the lender sets $f^*=0$ when he buys conditional on $b=0$ and the prices from Table 1. We then plug this cutoff into the prices and verify that all conjectured trades are optimal.

Let us look at the optimality of buying when $b=0$. Buying profits are still represented by (21), though now $E[P(z)|D_S+D_L=2] = \frac{1}{5}\left(1+3\frac{\rho}{\rho+(1-\rho)(1-G(c^*))}+\rho\right)$ and $E[P(z)|D_S+D_L=1] = \frac{1}{5}\left(3\frac{\rho}{\rho+(1-\rho)(1-G(c^*))}+\rho\right)$. Denoting $\eta^* \equiv \frac{\rho}{\rho+(1-\rho)(1-G(c^*))} \in [\rho,1]$, we can plug into (21) and rewrite the buying condition as

$$\eta^* > \frac{\eta^*}{5}(1+3\eta^*+\rho)+\frac{(1-\eta^*)}{5}(3\eta^*+\rho) \iff \eta^* > \rho$$

which will be true whenever $G(c^*) > 0$—as it is here. Thus, the lender will never deviate to not trading. We can also rule out deviating to selling by plugging into (22) but this time with $E[P(z)|D_S+D_L=0] = \frac{1}{5}\left(2\frac{\rho}{\rho+(1-\rho)(1-G(c^*))}+\rho\right)$ and $E[P(z)|D_S+D_L=-1] = \frac{1}{5}\left(\frac{\rho}{\rho+(1-\rho)(1-G(c^*))}+\rho\right)$. Following the same procedure as before, a necessary condition for positive selling profits is $\rho E[P(Z)|D_S+D_L=0]+(1-\rho)E[P(z)|D_S+D_L=-1]-\rho>0$. Plugging in, this expression equals $-\frac{\rho(\rho+5)}{5(\rho+2)}<0$, so deviating to selling is unprofitable for the lender.

Then, for the same reasons as the $D_L(b=0)=0$ case, the lender will also not want to deviate to any other fee plus trading strategy pair. □

Proof of Corollary 1. The first claim immediately follows from Table 1 for $z \in \{-4,-3,-2,-1,0,3,4\}$. Thus, to finish the proof of this claim, I must examine the cases where $z \in \{1,2\}$, which amounts to showing that c^* is no lower when $D_L^*(b=0)=0$ is the lender's trading strategy than when $D_L^*(b=0)=1$. From the proof of Proposition 3, we know that $c^* = \frac{2\rho}{5}$ when $D_L^*(b=0)=0$ and $c^* = \frac{\rho}{5}$ when $D_L^*(b=0)=1$, demonstrating the claim. The second claim about expected prices then follows from there being at least one value of z in every expected price (i.e., every expected price conditional on D_S+D_L for all $D_S+D_L \in \{-2,\dots,2\}$) where $P(z,D_L^*(b=0))$ is strictly lower when $D_L^*(b=0)=1$ than when $D_L^*(b=0)=0$. One such value is $z=-1$ for $D_S+D_L \in \{-2,\dots,1\}$ and $z=3$ for $D_S+D_L=2$. □

Proof of Lemma 3. To prove this claim, I first show that the lender's optimal fee violates the no-recall condition outlined in Eq. (11) when he buys or does not trade conditional on no loan agreement. From Eq. (10), these lenders' optimal fees are both $f^*=0$. Then, since $\Delta E[P(z)] = \frac{1}{5}(P(1)-P(-4))>0$, this optimal fee violates the no-recall condition. Next, observe that the no-recall condition is always satisfied when the lender sets the optimal selling fee $\frac{E[P(z)|D_S+D_L=-1]}{2}$. From (11), the condition will be satisfied if $\frac{E[P(z)|D_S+D_L=-1]}{2} > E[P(z)|D_S+D_L=-1]-E[P(z)|D_S+D_L=-2]$. By re-arranging, this amounts to $E[P(z)|D_S+D_L=-2]-\frac{E[P(z)|D_S+D_L=-1]}{2}>0$. Observe that this condition equals the shorting threshold for the speculator when $D_L(b=0)=-1$. So, if $c^*>0$, it must be satisfied. If $c^*=0$, the lender was already attracting zero short sellers, so increasing the fee to satisfy the condition has no impact on his profits. Thus, the lender will never need to deviate from his optimal fee when he sells following no loan agreement.

I must now show that the lender will choose $f^* = \Delta E[P(z)]$ when his optimal fee violates the no-recall constraint. I have already shown that the lender's expected profits are decreasing in f for all $f \in [0, E[P(z)|D_S+D_L=-2]]$ (and constant after that) when $D_L(b=0)=1$ in the proof of Proposition 2. As a result, this lender will choose the minimum fee that satisfies the no-recall constraint, $\Delta E[P(z)]$, as opposed to a fee that violates the constraint because such a fee is equivalent to an infinitely high fee (more formally, some $f > E[P(z)|D_S+D_L=-2]$) that eliminates all short selling.

The intuition is similar for lenders with $D_L(b=0)=0$ once I show that their profits are also strictly decreasing in f over the range $[0, E[P(z)|D_S+D_L=-2]$. First, note that $E[P(z)|D_S+D_L=-2]-\bar{C}<0$

again given our definition of $\bar{C}$. Then, for $D_L(b=0)=0$, the first derivative of expected profits with respect to f is $\frac{-2f(1-\rho)}{\bar{C}}$ over the range $f \in [0, E[P(z)|D_S+D_L=-2]$ and zero beyond that. Thus, the lender will choose the minimum fee satisfying the no-recall condition, $\Delta E[P(z)]$, since any fee in violation of the no-recall condition is again equivalent to a fee that eliminates all short selling. □

Proof of Proposition 4. I start by ruling out $D_L(b=0) \in \{-1,1\}$ as equilibrium trading strategies. First, note by Lemma 3, the no-recall constraint does not bind when $D_L(b=0)=-1$, so the proof for why this trade cannot occur in equilibrium is identical to that contained in the proof of Proposition 3.

Now I must rule out $D_L(b=0)=1$. The prices given this trading strategy are unchanged from Table 1, but now the equilibrium shorting threshold must satisfy the following fixed point problem

$$\begin{aligned} &E[P(z)|D_S+D_L=-2]-(E[P(z)|D_S+D_L=-1] \\ &\quad - E[P(z)|D_S+D_L=-2])=c^* \\ \iff &\rho-\frac{\rho}{\rho+(1-\rho)(1-G(c^*))}=5c^* \end{aligned} \tag{23}$$

First, note that $c^*=0$ is a solution to this problem. Now I must show that there are no other solutions. To see this, note that the left hand side is strictly decreasing in c^* while the right hand side is strictly increasing. Thus, for any other candidate $c^*>0$, the equality will not hold, meaning that $c^*=0$ is the unique solution (i.e., all short sales are eliminated). However, if all short sales are eliminated, the lending market signal is uninformative, which means that the lender becomes an uninformed trader. Thus, the lender cannot earn positive profits by buying. The intuition is that by raising fees, the lender's valuation conditional on $b=0$ declines (by Lemma 2), making buying unprofitable. We can see this formally by plugging the relevant prices into (21). Buying will be optimal whenever $\rho > \frac{1}{5}\rho(1+4\rho)+\frac{1}{5}(1-\rho)(4\rho)$; however, this never holds since instead $\rho=\frac{1}{5}\rho(1+4\rho)+\frac{1}{5}(1-\rho)(4\rho)$, meaning that the lender never earns strictly positive profits by buying, so he would rather not trade.

Now I address the remaining possible pure strategy trade $D_L(b=0)=0$. Again, prices are as described in Table 1, but now c^* solves

$$\begin{aligned} &E[P(z)|D_S+D_L=-2]-(E[P(z)|D_S+D_L=-1] \\ &\quad - E[P(z)|D_S+D_L=-2])=c^* \\ \iff &2\rho-\frac{\rho}{\rho+(1-\rho)(1-G(c^*))}=5c^* \end{aligned}$$

Only the lower root of this quadratic equation solves the model: $c^*=\frac{5\bar{C}+2\rho-\sqrt{25\bar{C}^2+4\rho^4-8\rho^3+4\rho^2}-2\rho^2}{10(1-\rho)}$. To show that the solution is interior, observe that

$$c^* < \frac{5\bar{C}+2\rho-\sqrt{25\bar{C}^2}-2\rho^2}{10(1-\rho)}=\frac{\rho}{5}<\bar{C} \tag{24}$$

$$\text{since } 4\rho^4-8\rho^3+4\rho^2>0 \text{ for all } \rho\in(0,1)$$

by the definition of $\bar{C}=\frac{3\rho}{5}$ in Assumption 2 and Proposition S1. The other root of the quadratic is not a solution because it is always greater than $\bar{C}$ and $c^*=\bar{C}$ is not an equilibrium. That $c^*=\bar{C}$ is not an equilibrium follows immediately from the equilibrium condition $2\rho-\frac{\rho}{\rho+(1-\rho)(1-G(c^*))}=5c^*$ from above; plugging in, $\bar{C}$ will be an equilibrium if $\frac{2\rho-1}{5}>\bar{C}$ which never occurs by the definition of $\bar{C}$. Next, the other root being greater than $\bar{C}$ can be shown by noting that

$$\begin{aligned} &\frac{5\bar{C}+2\rho+\sqrt{25\bar{C}^2+4\rho^4-8\rho^3+4\rho^2}-2\rho^2}{10(1-\rho)}>\bar{C} \\ \iff &5\bar{C}+2\rho+\sqrt{25\bar{C}^2+4\rho^4-8\rho^3+4\rho^2}-2\rho^2>10(1-\rho)\bar{C} \end{aligned}$$

Next observe that

$$5\bar{C}+2\rho+\sqrt{25\bar{C}^2+4\rho^4-8\rho^3+4\rho^2}-2\rho^2>10\bar{C}+2\rho(1-\rho)>10(1-\rho)\bar{C}$$

since $4\rho^4-8\rho^3+4\rho^2>0$.

Now I must check whether the lender wants to deviate. To prove that the lender will not deviate to buying, we can plug in and rearrange (21) to show that buying profits will be negative whenever $c^*<\bar{C}\frac{1+\rho}{1+2\rho}$. Then, to show that this inequality holds, we can first recall from (24) that $c^*<\frac{\rho}{5}$. Next, note that $\frac{\rho}{5}<\bar{C}\frac{1+\rho}{1+2\rho}$ when $\bar{C}=\frac{3\rho}{5}$ as defined in Assumption 2 and Proposition S1. Thus, it must also be true that $c^*<\bar{C}\frac{1+\rho}{1+2\rho}$, meaning that the lender will not deviate to buy.

Next, to prove that the will not deviate to selling, recall that a necessary condition for positive deviant selling profits is $\rho E[P(z)|D-H+D_L=0]+(1-\rho)E[P(z)|D_S+D_L=-1]-\rho>0$ (from (22)). Plugging in, this expression equals $\frac{3\rho\left(3\rho+\rho\sqrt{4\rho^2-8\rho+13}-2\rho^2-6\right)}{5\left(\sqrt{4\rho^2-8\rho+13}-4\rho+7\right)}<0$ for all $\rho\in(0,1)$. Thus, the lender will not deviate to selling at any deviant fee.

Similar to the proof of Proposition 3, the lender will also not deviate to any other fee and trading strategy pair since we already ruled out deviations to the optimal buying fee, and the proof ruling out selling is valid for all possible fees. □

Proof of Corollary 2. From Proposition 4, the active lender's equilibrium trading strategy conditional on no loan agreement is $D_L^*(b=0)=0$, and from the proof of Lemma 3, we know the active lender's expected profits are decreasing in the fee for all $f>0$. Thus, when this active lender is forced by recall risk to increase his fee from 0 to $\Delta E[P(z)]$, his expected profits must decline. □

Proof of Proposition 5. I start by showing that the passive lender always sets a higher fee. Denote f_p^* as the passive lender's fee and f_a^* as the active lender's fee. This claim is trivial when there is no recall risk because $f_p^*=\frac{3\rho}{10}>0=f_a^*$. Now suppose there is recall risk. Then

$$\begin{aligned} f_p^*>f_a^* &\iff \frac{3\rho}{10}>\frac{\rho}{5(\rho+(1-\rho)(1-G(c_a^*)))} \\ &\iff 3>\frac{2}{\rho+(1-\rho)(1-G(c_a^*))} \\ &\iff 1>3(1-\rho)G(c_a^*) \\ &\iff 1>\frac{5(1-\rho)c_a^*}{\rho} \\ &\iff \frac{\rho}{5}>c_a^*(1-\rho) \end{aligned}$$

where c_a^* is the equilibrium shorting threshold under an active lender and the fourth line follows from $\bar{C}=\frac{3\rho}{5}$ (by Assumption 2). The claim then immediately follows because we know $c_a^*<\frac{\rho}{5}$ from the proof of Proposition 4.

I now compare equilibrium shorting thresholds. This proof amounts to comparing the equilibrium values of c^* found with an active lender to those of the passive lender from Propositions 1, 3, and 4. Greater values of c^* indicate more equilibrium shorting.

From Proposition 1, $c^*=\frac{3\rho}{10}$ with a passive lender. From Proposition 3, $c^*=\frac{\rho}{5}$ when $D_L^*(b=0)=1$ and $c^*=\frac{2\rho}{5}$ when $D_L^*(b=0)=0$. Thus, without recall risk, the passive lender attracts more short selling than the active lender if and only if $D_L^*(b=0)=1$.

From Proposition 4, $c^*=\frac{5\bar{C}+2\rho-\sqrt{25\bar{C}^2+4\rho^4-8\rho^3+4\rho^2}-2\rho^2}{10(1-\rho)}$ with an active lender. Recall from the proof of Proposition 4 that $\frac{5\bar{C}+2\rho-\sqrt{25\bar{C}^2+4\rho^4-8\rho^3+4\rho^2}-2\rho^2}{10(1-\rho)}<\frac{\rho}{5}<\frac{3\rho}{10}$, so with recall risk, the passive lender attracts greater shorting than the active lender. □

Proof of Proposition 6. Price efficiency is defined as $1-E[|v-E[P(z)|v]|]=1-\rho+\rho E[P(z)|v=1]-(1-\rho)E[P(z)|v=0]$

First, I start with the passive lender. For the passive lender, $E[P(z)|v=1]=E[P(z)|D_S+D_L=1]$ and $E[P(z)|v=0]=G(c^*)E[P(z)|D_S+D_L=-1]+(1-G(c^*))E[P(z)|D_S+D_L=0]$. Then, using the prices from Eq. (2) in the main text and the result from

Proposition 1 that $c^* = \frac{3\rho}{10}$, we get the following measure of price efficiency: $Eff_p = \frac{50\overline{C}-15\rho-80\overline{C}\rho+80\overline{C}\rho^2+39\rho^2-42\rho^3+18\rho^4}{5\left(3\rho^2-3\rho+10\overline{C}\right)}$.

Now I address the active lender without recall risk. When the active lender buys conditional on $b = 0$, the expected prices are $E[P(z)|v = 1] = E[P(z)|D_S + D_L = 2]$ and $E[P(z)|v = 0] = G(c^*)E[P(z)|D_S + D_L = -2] + (1 - G(c^*))E[P(z)|D_S + D_L = 1]$, and the cutoff is $c^* = \frac{\rho}{5}$. Plugging into the prices from Table 1 we get the following measure of price efficiency $Eff_a(NR, 1) = \frac{25\overline{C}-5\rho-40\overline{C}\rho+40\overline{C}\rho^2+13\rho^2-10\rho^3+2\rho^4}{5\left(\rho^2-\rho+5\overline{C}\right)}$. When the active lender does not trade conditional on $b = 0$, the expected prices are $E[P(z)|v = 1] = E[P(z)|D_S + D_L = 1]$ and $E[P(z)|v = 0] = G(c^*)E[P(z)|D_S + D_L = -2] + (1 - G(c^*))E[P(z)|D_S + D_L = 0]$, and the cutoff is $c^* = \frac{2\rho}{5}$. Plugging again into the prices from Table 1 we get the following measure of price efficiency $Eff_a(NR, 0) = \frac{25\overline{C}-10\rho-40\overline{C}\rho+40\overline{C}\rho^2+26\rho^2-24\rho^3+8\rho^4}{5\left(2\rho^2-2\rho+5\overline{C}\right)}$.

For the active lender with recall risk, the expected prices are $E[P(z)|v = 1] = E[P(z)|D_S + D_L = 1]$ and $E[P(z)|v = 0] = G(c^*)E[P(z)|D_S + D_L = -2] + (1 - G(c^*))E[P(z)|D_S + D_L = 0]$, and the cutoff is $c^* = \frac{5\overline{C}+2\rho-\sqrt{25\overline{C}^2+4\rho^4-8\rho^3+4\rho^2-2\rho^2}}{10(1-\rho)}$. Plugging into the prices from Table 1 we get the following measure of price efficiency

$$Eff_a(R) = \frac{25\overline{C} - 10\rho - 40\overline{C}\rho - 8\rho\phi + 5\phi + 60\overline{C}\rho^2 + 4\rho^2\phi + 26\rho^2 - 24\rho^3 + 8\rho^4}{5\left(5\overline{C} - 2\rho + \phi + 2\rho^2\right)}$$

where $\phi \equiv \sqrt{25\overline{C}^2 + 4\rho^4 - 8\rho^3 + 4\rho^2}$

First, $Eff_p - Eff_a(NR, 1) = -\frac{6\rho^3(1-\rho)\left(2\rho^2-2\rho+5\overline{C}\right)}{5\left(50\overline{C}^2+25\overline{C}\rho^2-25\overline{C}\rho+3\rho^4-6\rho^3+3\rho^2\right)} = -\frac{6\rho^3(2\rho^2+\rho)(1-\rho)}{5(3\rho^4+9\rho^3+6\rho^2)} < 0$ when $\bar{C} = \frac{3\rho}{5}$ as defined in Assumption 2 and Proposition S1. Now observe that similarly, $Eff_p - Eff_a(NR, 0) = -\frac{2\rho^3(1-\rho)\left(6\rho^2-6\rho+25\overline{C}\right)}{5\left(50\overline{C}^2+35\overline{C}\rho^2-35\overline{C}\rho+6\rho^4-12\rho^3+6\rho^2\right)} = -\frac{2\rho^3(6\rho^2+9\rho)(1-\rho)}{5(6\rho^4+9\rho^3+3\rho^2)} < 0$ again when $\bar{C} = \frac{3\rho}{5}$ as defined in Assumption 2 and Proposition S1. Thus, passive efficiency is always lower when there is no recall risk.

Finally,

$$Eff_p - Eff_a(R) = \frac{2\rho^2\left(5\overline{C}\rho-3\rho\phi-5\overline{C}\rho^2+3\rho^2\phi-100\overline{C}^2+6\rho^2-12\rho^3+6\rho^4+20\overline{C}\phi\right)}{5\left(3\rho^2-3\rho+10\overline{C}\right)\left(5\overline{C}-2\rho+\phi+2\rho^2\right)}$$

where again $\phi \equiv \sqrt{25\overline{C}^2 + 4\rho^4 - 8\rho^3 + 4\rho^2}$. Plugging in $\bar{C} = \frac{3\rho}{5}$ then gives us

$$Eff_p - Eff_a(R) = \frac{2\rho^2\left(9\rho\sqrt{4\rho^4-8\rho^3+13\rho^2}+3\rho^2\sqrt{4\rho^4-8\rho^3+13\rho^2}-27\rho^2-15\rho^3+6\rho^4\right)}{5(3\rho^2+3\rho)\left(\rho+\sqrt{4\rho^4-8\rho^3+13\rho^2}+2\rho^2\right)}$$

which will be positive if and only if $9\rho\sqrt{4\rho^4 - 8\rho^3 + 13\rho^2} + 3\rho^2\sqrt{4\rho^4 - 8\rho^3 + 13\rho^2} + 6\rho^4 > 27\rho^2 + 15\rho^3$ which will hold for all $\rho \in (0, \frac{\sqrt{13}-2}{3})$. Thus, passive efficiency can be higher than active efficiency with recall risk when ρ is sufficiently small. □

Proofs of Section 4

Proof of Lemma 4. This proof amounts to showing that, given some trading strategy, the lender's private signal does not affect his optimal fee choice as a function the prices.[43] The lender's objective functions are

If $D_L^y(b = 0) = 1$:

$$Pr[v = 0 \mid y]G(c_y^*)(E[P(z) \mid D_S + D_L = -2] + f)$$
$$+ Pr[v = 1 \mid y](1 - E[P(z) \mid D_S + D_L = 2])$$
$$+ Pr[v = 0 \mid y](1 - G(c_y^*))(0 - E[P(z) \mid D_S + D_L = 1])$$

If $D_L^y(b = 0) = 0$:

$$Pr[v = 0 \mid y]G(c_y^*)(E[P(z) \mid D_S + D_L = -2] + f)$$

If $D_L^y(b = 0) = -1$:

$$Pr[v = 0 \mid y]G(c_y^*)(E[P(z) \mid D_S + D_L = -2] + f)$$
$$+ Pr[v = 1 \mid y](E[P(z) \mid D_S + D_L = 0] - 1)$$
$$+ Pr[v = 0 \mid y](1 - G(c_y^*))(E[P(z) \mid D_S + D_L = -1] - 0)$$

where $G(c_y^*) = \frac{E[P(z)|D_S+D_L=-2]-f}{\bar{C}}$ and $Pr[v = 0|y]$, $Pr[v = 1|y]$ are functions of q and ρ. The first term in each objective function corresponds to the event where a loan occurs, the second term corresponds to no loan occurring and $v = 1$, and the third term corresponds to no loan occurring and $v = 0$.[44] Observe that the fee only appears in the first and third terms of these objective functions (directly though f and indirectly through $G(c_y^*)$). Thus, in the lender's first-order condition, $Pr[v = 0|y]$ cancels out (since it is a multiplicative constant) and $Pr[v = 1 \mid y]$ does not appear, meaning that q does not affect the lender's optimal fee as a function of the equilibrium prices. It then follows that the lender's optimal fees will continue to be described by Eq. (10) from the main text and that the no-recall condition will only bind when $D_L^y(b = 0) \in \{0, 1\}$, leading the lender to set the minimum fee satisfying this condition: $\Delta E[P(z)] > 0$. □

Proof of Corollary 3. Lemma 1 is still valid, so the lender continues to sell following a loan agreement regardless of his private signal. Moreover, the unique equilibrium trade conditional on no loan agreement is for the positively informed L_1 to buy and the negatively informed L_0 to sell. This outcome follows because L_1's valuation is arbitrarily close to one and the expected buying price is always strictly less than one. The intuition for L_0 selling is analogous. Thus, by Lemma 4, L_1 sets $f_1^* = 0$ without recall risk and $f_1^* = \Delta E[P(z)]$ with recall risk, while L_0 always chooses $f_0^* = \frac{E[P(z)|D_S+D_L=-1]}{2}$. Without recall risk, f_0^* is clearly greater than f_1^* since expected prices are always strictly positive. With recall risk, the condition that determines whether $f_0^* > f_1^*$ is simply

$$\begin{aligned} f_0^* > f_1^* &\iff \frac{E[P(z)|D_S + D_L = -1]}{2} \\ &> E[P(z)|D_S + D_L = -1] - E[P(z)|D_S + D_L = -2] \\ &\iff E[P(z)|D_S + D_L = -2] - \frac{E[P(z)|D_S + D_L = -1]}{2} > 0 \\ &\iff c_0^* > 0 \end{aligned}$$

completing the final part of the corollary. □

Appendix B. Supplementary data

Supplementary material related to this article can be found online at https://doi.org/10.1016/j.jfineco.2026.104237.

Data availability

Learning by Lending Securities code (Original data) (Mendeley Data)

References

Atmaz, A., Basak, S., Ruan, F., 2023. Dynamic equilibrium with costly short-selling and lending market. Rev. Financ. Stud. 37 (2), 444–506. http://dx.doi.org/10.1093/rfs/hhad060.

Bagwell, K., Ramey, G., 1991. Oligopoly limit pricing. Rand J. Econ. 22 (2), 155–172, URL: http://www.jstor.org/stable/2601015.

Baldauf, M., Mollner, J., 2024. Competition and information leakage. J. Political Econ. 132 (5), 1603–1641. http://dx.doi.org/10.1086/727709.

Baruch, S., Zhang, X., 2022. The distortion in prices due to passive investing. Manag. Sci. 68 (8), 6219–6234. http://dx.doi.org/10.1287/mnsc.2021.4114.

[43] Of course, the private signal does affect what trading strategy the lender ultimately chooses and what those prices equal.

[44] The last two terms equal zero when the lender does not trade conditional on $b = 0$.

von Beschwitz, B., Honkanen, P., Schmidt, D., 2025. Passive ownership and short selling. Rev. Financ. 29 (4), 1137–1188. http://dx.doi.org/10.1093/rof/rfaf023.

Blocher, J., Reed, A.V., Van Wesep, E.D., 2013. Connecting two markets: An equilibrium framework for shorts, longs, and stock loans. J. Financ. Econ. 108 (2), 302–322. http://dx.doi.org/10.1016/j.jfineco.2012.12.006.

Blocher, J., Whaley, R., 2016. Two-sided markets in asset management: Exchange-traded funds and securities lending. Work. Pap. URL: https://papers.ssrn.com/sol3/Delivery.cfm?abstractid=2474904.

Bond, P., Garcia, D., 2022. The equilibrium consequences of indexing. Rev. Financ. Stud. 35 (7), 3175–3230. http://dx.doi.org/10.1093/rfs/hhab106.

Brown, D.C., Davies, S.W., 2017. Moral hazard in active asset management. J. Financ. Econ. 125 (2), 311–325. http://dx.doi.org/10.1016/j.jfineco.2017.05.010.

Buffa, A.M., Vayanos, D., Woolley, P., 2022. Asset management contracts and equilibrium prices. J. Political Econ. 130 (12), 3146–3201. http://dx.doi.org/10.1086/720515.

Cen, X., Dou, W.W., Kogan, L., Wu, W., 2024. Fund flows and income risk of fund managers. http://dx.doi.org/10.3386/w31986, NBER Working Paper 31986.

Chague, F.D., Giovannetti, B., Herskovic, B., 2025. Information leakage from short sellers. http://dx.doi.org/10.3386/w31927, NBER Working Paper 31927.

Chen, S., Kaniel, R., Opp, C., 2025. Market power in the securities lending market. http://dx.doi.org/10.2139/ssrn.4100699, Available at SSRN.

Chinco, A., Sammon, M., 2024. The passive ownership share is double what you think it is. J. Financ. Econ. 157, 103860. http://dx.doi.org/10.1016/j.jfineco.2024.103860.

Christoffersen, S.E., Geczy, C.C., Musto, D.K., Reed, A.V., 2007. Vote trading and information aggregation. J. Financ. 62 (6), 2897–2929. http://dx.doi.org/10.1111/j.1540-6261.2007.01296.x.

Chuprinin, O., Ruf, T., 2018. Let the bear beware: What drives stock recalls. http://dx.doi.org/10.2139/ssrn.2831261, Available at SSRN.

Cohen, L., Diether, K.B., Malloy, C.J., 2007. Supply and demand shifts in the shorting market. J. Financ. 62 (5), 2061–2096. http://dx.doi.org/10.1111/j.1540-6261.2007.01269.x.

Coles, J.L., Heath, D., Ringgenberg, M.C., 2022. On index investing. J. Financ. Econ. 145 (3), 665–683. http://dx.doi.org/10.1016/j.jfineco.2022.05.007.

Duffie, D., Gârleanu, N., Pedersen, L.H., 2002. Securities lending, shorting, and pricing. J. Financ. Econ. 66 (2), 307–339. http://dx.doi.org/10.1016/S0304-405X(02)00226-X.

Duong, T., Huszar, Z., Tan, R., Zhang, W., 2017. The information value of stock lending fees: Are lenders price takers? Rev. Financ. 21 (6), 2353–2377. http://dx.doi.org/10.1093/rof/rfw075.

Engelberg, J.E., Reed, A.V., Ringgenberg, M.C., 2018. Short-selling risk. J. Financ. 73 (2), 755–786. http://dx.doi.org/10.1111/jofi.12601.

Greppmair, S., Jank, S., Saffi, P., Sturgess, J., 2024. Securities lending and information acquisition. http://dx.doi.org/10.2139/ssrn.3604977, Available at SSRN.

Honkanen, P., 2025. Securities lending and trading by active and passive funds. J. Financ. Quant. Anal. 60 (3), 1272–1309. http://dx.doi.org/10.1017/S0022109024000966.

Ibert, M., Kaniel, R., Van Nieuwerburgh, S., Vestman, R., 2017. Are mutual fund managers paid for investment skill? Rev. Financ. Stud. 31 (2), 715–772. http://dx.doi.org/10.1093/rfs/hhx105.

Johnson, T.L., Weitzner, G., 2025. Distortions caused by lending fee retention. Manag. Sci. 71 (1), 35–58. http://dx.doi.org/10.1287/mnsc.2021.01390.

Jones, C., Lamont, O., 2002. Short-sale constraints and stock returns? J. Financ. Econ. 66 (2–3), 207–239. http://dx.doi.org/10.1016/S0304-405X(02)00224-6.

Kashyap, A.K., Kovrijnykh, N., Li, J., Pavlova, A., 2023. Is there too much benchmarking in asset management? Am. Econ. Rev. 113 (4), 1112–1141. http://dx.doi.org/10.1257/aer.20210476.

Kolasinski, A., Reed, A., Ringgenberg, M., 2013. A multiple lender approach to understanding supply and search in the equity lending market. J. Financ. 68 (2), 559–595. http://dx.doi.org/10.1111/jofi.12007.

Kyle, A.S., 1985. Continuous auctions and insider trading. Econometrica 53 (6), 1315–1335, URL: http://www.jstor.org/stable/1913210.

Lamont, O.A., 2012. Go down fighting: Short sellers vs. Firms. Rev. Asset Pricing Stud. 2 (1), 1–30. http://dx.doi.org/10.1093/rapstu/ras003.

Ljungqvist, A., Qian, W., 2016. How constraining are limits to arbitrage? Rev. Financ. Stud. 29 (8), 1975–2028. http://dx.doi.org/10.1093/rfs/hhw028.

Mainardi, F., 2023. A demand-based approach to short-selling. http://dx.doi.org/10.2139/ssrn.4418398, Available at SSRN.

Nezafat, M., Schroder, M., 2021. Private information, securities lending, and asset prices. Rev. Financ. Stud. 35 (2), 1009–1063. http://dx.doi.org/10.1093/rfs/hhab043.

Nurisso, G., 2025. Paying for peer performance: index investing on managerial incentives. http://dx.doi.org/10.2139/ssrn.5258647, Available at SSRN.

Palia, D., Sokolinski, S., 2024. Strategic borrowing from passive investors. Rev. Financ. 28 (5), 1537–1573. http://dx.doi.org/10.1093/rof/rfae012.

Pankratov, A., 2020. Securities lending and information transmission: A model of endogenous short-sale constraints. http://dx.doi.org/10.2139/ssrn.3680016, Available at SSRN.

Reed, A.V., 2013. Short selling. Annu. Rev. Financ. Econ. 5, 245–258. http://dx.doi.org/10.1146/annurev-financial-110311-101813.

Sikorskaya, T., 2025. Institutional investor mandates, securities lending, and short-selling constraints. http://dx.doi.org/10.2139/ssrn.4627075, Available at SSRN.